# SOCRATES

# User's Manual




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# Chapter 1

# Introduction

## 1.1 Overview

SOCRATES (Signals of Opportunity Constellation and Remote sensing Analysis Tool for Earth Science) is an end-to-end mission simulator for multiple Signals of Opportunity (multi-SoOp) for land remote sensing. This tool offers a comprehensive analysis environment to explore from a high-level mission design to scientific measurements. The current release of SOCRATES boasts the following capabilities:

- Satellite- and tower-based observation of soil moisture using multi-SoOp.
- Tradespace exploration of multi-SoOp constellation design.
- Orbit propagation and import of multiple receivers and transmitters.
- Specular reflection point prediction on the Earth surface.
- Instrument property realizations including channel frequency, bandwidth, coherent integration time, and noise temperature.
- Antenna property realizations including antenna orientation and gain pattern.
- Geophysical data realizations including soil moisture, soil temperature, soil texture, vegetation, land cover, freeze/thaw state, and snow cover.
- Link budget for direct and reflected signals.
- Modeling of multi-body spacecraft attitude dynamics with rigid bodies.

Use **SOCRATES-Retrieval** for end-to-end retrieval analysis using multi-SoOp. SOCRATES-Retrieval can be accessed from the following Code Ocean capsule: `https://doi.org/10.24433/CO.5405959.v2`.

## 1.2 Compilation

The GNU Compiler Collection with C11 standard on a Unix or Linux platform is recommended to compile the code. Additional software packages OpenMPI and netCDF are required for the compilation. The latest versions of the packages are recommended. OpenMPI is used for multi-processing the mission design processes. NetCDF is for creating, reading, and writing the NetCDF files.

SOCRATES can be accessed from the following GitHub repository: `https://github.itap.purdue.edu/RadioNavigationLab/SOCRATES`. To install from the GitHub repository:

1. Download the source code from the latest release or clone the code from GitHub:

   ```
   $ git clone git@github.itap.purdue.edu:RadioNavigationLab/SOCRATES.git
   ```

2. Open a terminal on the top-level directory (i.e., **.../SOCRATES**) and run `$ make` to compile the simulator. Note that `$ make clean` deletes the existing object files and the executable file.

3. An executable file **SOCRATES** should be created in the top-level directory.

## 1.3 Downloading Large Binaries

In the current version, the following large files must exist in each corresponding directory:

- A 1-km resolution land mask: **./earthdata/static/Land_Mask_1km_EASE2.uint8**

These files are distributed through releases. To download:

1. Visit `https://github.itap.purdue.edu/RadioNavigationLab/SOCRATES/releases` and click the latest release.

2. Download the necessary large files attached to the release.

3. Place each file in each corresponding directory.

## 1.4 Getting Started

SOCRATES is **a terminal-based application** where four analyses are available: *Satellite-based observation*, *Tower-base observation*, *Fixed observation*, and *Mission design*. It also provides a MATLAB-based Graphical User Interface (GUI) for easy setup and interactive analysis. For satellite-based observation and mission design, run **SOCRATES.mlapp** in the top-level directory and follow the instructions in Chapter 3 and Chapter 6, respectively. All simulation projects should be located in **./sim** folder. It includes a demo project for each analysis type:

- **Demo-space**: A demo project for *satellite-based observation* analysis. See Chapter 3 for a step-by-step guide to running the simulation.

- **Demo-tower**: A demo project for *tower-based observation* analysis. See Chapter 4 for a step-by-step guide to running the simulation.

- **Demo-fixed**: A demo project for *fixed observation* analysis. See Chapter 5 for a step-by-step guide to running the simulation.

- **Demo-design**: A demo project for *mission design* analysis. See Chapter 6 for a step-by-step guide to running the simulation.

Each project folder should contain input text files required by the simulator. **Inp_Main.txt** is the top-level input file and outputs will be stored in **.sim/[project name]/out** directory. Details on input and output files are described in Chapter 7.

To run the SOCRATES on a terminal, open a terminal in the top-level directory and enter `$ SOCRATES` to run the simulation. The default project is **Demo-space**, if no argument is specified. The project to run can be changed by running the SOCRATES with an argument:

```
$ SOCRATES [project name]
```

## 1.5 Citation

SOCRATES is open-source software under GNU General Public Licence (GPL) and is freely available with its documentation. However, we would appreciate those who cite the corresponding studies below if they are used:

SOCRATES:

Seho Kim and James Garrison, "A Mission Design Tool for Satellite Constellations Using Multi-Frequency Signals of Opportunity," IGARSS 2023 - 2023 IEEE International Geoscience and Remote Sensing Symposium, 2023.

For any questions, please contact Seho Kim (kim3002@purdue.edu).

## 1.6 Disclosure of Third-party Code

This software includes material copied from or derived from the following sources:

- Orbit propagator from 42:
  https://github.com/ericstoneking/42, Copyright (C) 2010 United States Government as represented by the Administrator of the National Aeronautics and Space Administration.
- The Principle of Maximum Entropy (POME) model translated from Python code:
  https://github.com/Vikalp86/soil-moisture-profile, Copyright (C) 2019 Vikalp86.
- The SCoBi model translated from MATLAB code:
  https://github.com/impresslab/SCoBi, Copyright (C) 2017-2018 Mehmet Kurum, Orhan Eroglu, Dylan R. Boyd.

# Chapter 2

# SOCRATES Basics

## 2.1 Analysis Type

The SOCRATES provides four types of analysis:

- *Satellite-based observation*: The satellite-based observation simulates SoOp observation in space. Multiple receivers can be placed in any orbit. It provides not only SoOp observables based on the SCoBi model but also various parameters that come from a bistatic SoOp observation.
- *Tower-based observation*: The tower-based observation simulates SoOp observation on the ground. It is similar to the satellite-based observation but a single receiver is mounted on a fixed tower. Transmitters are in their orbits.
- *Fixed observation*: The fixed observation generates outputs of the SCoBi forward scattering model, assuming the locations of both a transmitter and a receiver are fixed.
- *Mission desgin*: The mission designer provides a framework to explore the tradespace of a multi-SoOp constellation and help find the optimal design upon the user's preference.

## 2.2 Main Window

The user can select an analysis type on the main window shown in Figure 2.1. This window is designed to provide an easy way of selecting the analysis type and configuring input for analysis. The SOCRATES comes with an initial set of default inputs for each analysis type with a number of SoOp configurations with various signal sources. The GUI is offered for the following analyses:

- Satellite-based Observation
- Mission Design

The following analyses do not support GUI, but they are currently available in the terminal:

- Tower-based Observation
- Fixed Observation

## 2.3 Orbit Propagator

The SOCRATES adopts the orbit and attitude propagation model of "42" simulator [1]. However, it supports importing satellite states (position and velocity) along with the modified Julian day in a text file. The orbit input files are located in **./orbit**. The user can select the orbit propagation model in **Inp_Inorbit.txt** or from GUI.

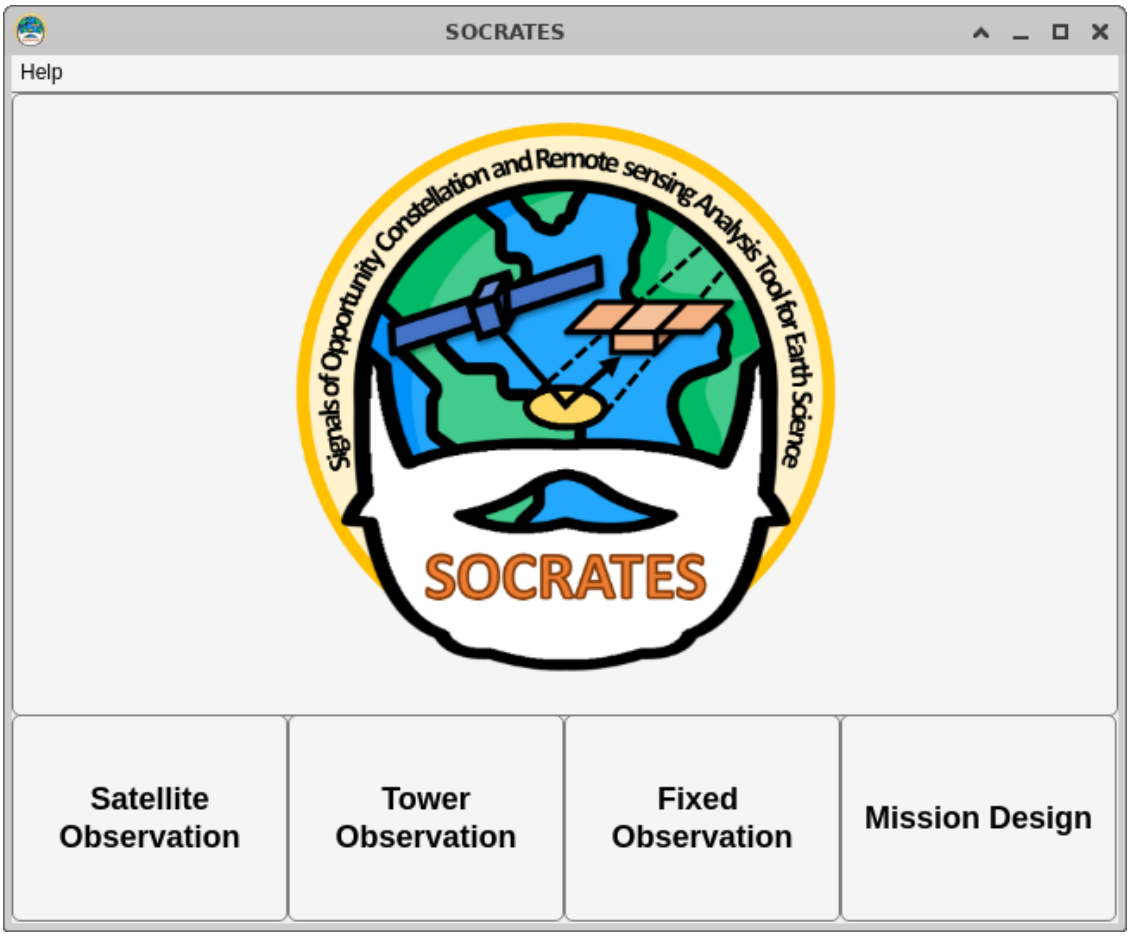



Figure 2.1: SOCRATES Main Window

### 2.3.1 42 Orbit Propagator

The orbit and attitude propagation model of the 42 simulator provides the position and velocity of satellites used in the SOCRATES. The original 42 model was modified to account for the orbital decay caused by atmospheric drag by applying the orbit rectification and Schatten solar flux model. The 42 is located in **./orbit/42** and the inputs for the 42 are located in **./orbit/42input**. The top-level input file for the 42 is **Inp_Sim.txt**. The input parameters in **Inp_Sim.txt** are described in Chapter 7.

### 2.3.2 Importing Orbit

The existing satellite states can be used in the SOCRATES. The user must provide the satellite states of all satellites in separate text files. The required data are the modified Julian day, the position vector (x, y, z) and the velocity vector (x, y, z) in the Earth-centered inertial (ECI) coordinate system in meters, separated by spaces. The text file should include a header in the first line.

# Chapter 3

# Satellite-based Observation

The provided simulation project for the space-based observation is a single CYGNSS observatory as an example of GNSS-R. The transmitting system is GPS consisting of 31 satellites in a medium Earth orbit (MEO). The observatory is in a low Earth orbit (LEO) at 520 km altitude and 35° inclination. The maximum number of specular points simultaneously trackable by the observatory is 4. The observatory is a 12-U CubeSat. Note that this spacecraft model is not consistent with the actual CYGNSS observatory. The SOCRATES will determine the specular reflection points and generate outputs that are requested in **Inp_Report.txt**. The simulation duration is 2 hours.

The satellite-based observation analysis can be executed in a terminal or using GUI.

## 3.1 Run in Terminal

1. The included input files are configured for the space-based observation by default. If the user wants to see and modify output parameters, open the **Inp_Report.txt** located in **./sim/Demo-space** folder and select output parameters of interest. `TRUE`/`FALSE` turns on/off the corresponding parameter.

2. Open the command prompt at the top-level directory and enter the command:

   ```
   $ SOCRATES Demo-space
   ```

   It will load the simulation inputs located in the **./sim/Demo-space** folder.

```
kim3002@cadia:~/Desktop/SOCRATES$ SOCRATES Demo-space
>> Loading land mask ...Success!
0.0 SC[0] Cmd Angles = [90.0 0.0 0.0] deg, Seq = 321 wrt L Frame
0.0 SC[1] Cmd Angles = [90.0 0.0 0.0] deg, Seq = 321 wrt L Frame
0.0 SC[2] Cmd Angles = [90.0 0.0 0.0] deg, Seq = 321 wrt L Frame
0.0 SC[3] Cmd Angles = [90.0 0.0 0.0] deg, Seq = 321 wrt L Frame
0.0 SC[4] Cmd Angles = [90.0 0.0 0.0] deg, Seq = 321 wrt L Frame
0.0 SC[5] Cmd Angles = [90.0 0.0 0.0] deg, Seq = 321 wrt L Frame
0.0 SC[6] Cmd Angles = [90.0 0.0 0.0] deg, Seq = 321 wrt L Frame
0.0 SC[7] Cmd Angles = [90.0 0.0 0.0] deg, Seq = 321 wrt L Frame
Reached CmdScript EOF at Time = 0.000000
/*************************************************************************/
/*                                                                       */
/*                              SOCRATES                                 */
/*                                                                       */
/*  SoOp Constellation and Remote sensing Analysis Tool for Earth Science  */
/*                                                                       */
/*************************************************************************/
>> Number of Input Soil Layers is 1
>> Initiate In-Orbit Observation ...
>> - Number of Receivers : 1
>> - Number of Transmitter Systems : 1
>> - Orbit Propagator : 42
    42 Case ./orbit/42input/Demo-space/ is   0% Complete at Time =        0.000
    42 Case ./orbit/42input/Demo-space/ is  10% Complete at Time =      720.000
    42 Case ./orbit/42input/Demo-space/ is  20% Complete at Time =     1440.000
    42 Case ./orbit/42input/Demo-space/ is  30% Complete at Time =     2160.000
    42 Case ./orbit/42input/Demo-space/ is  40% Complete at Time =     2880.000
    42 Case ./orbit/42input/Demo-space/ is  50% Complete at Time =     3600.000
    42 Case ./orbit/42input/Demo-space/ is  60% Complete at Time =     4320.000
    42 Case ./orbit/42input/Demo-space/ is  70% Complete at Time =     5040.000
    42 Case ./orbit/42input/Demo-space/ is  80% Complete at Time =     5760.000
    42 Case ./orbit/42input/Demo-space/ is  90% Complete at Time =     6480.000
    42 Case ./orbit/42input/Demo-space/ is 100% Complete at Time =     7200.000
     Total Run Time =      6.00 sec
     Sim Speed =  1200.00 x Real
>> Simulation End.
kim3002@cadia:~/Desktop/SOCRATES$
```

Figure 3.1: Execution screenshot of satellite-based observation demo

3. After completion, the output files will be located in the **./sim/Demo-space/out** folder. Figure 3.2 shows the example of output files. See Section 7.4 for the naming convention and description of the output file.




Figure 3.2: Products of satellite-based observation

## 3.2 Run using GUI

The user can utilize a GUI for satellite-based observation. For this, run **SOCRATES.mlapp** located on the top-level directory. The current folder of MATLAB must be set to the top-level directory of the SOCRATES (i.e. **./**). From the SOCRATES main window, click the "Satellite Observation" button to select the satellite-based observation analysis. This will open a path selector as shown in Figure 3.3. Select **Demo-space** folder to open the demo project and click the "Open" button. The user must select a project configured for satellite-based observation. This is set in **Inp_Main.txt**.

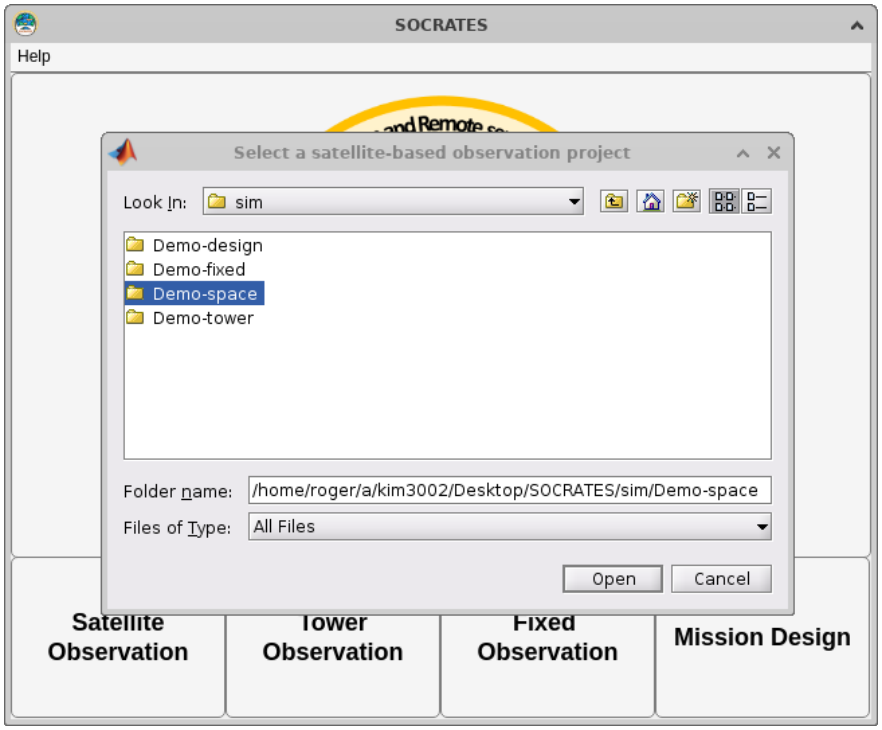



Figure 3.3: Project selection window for satellite-based observation.

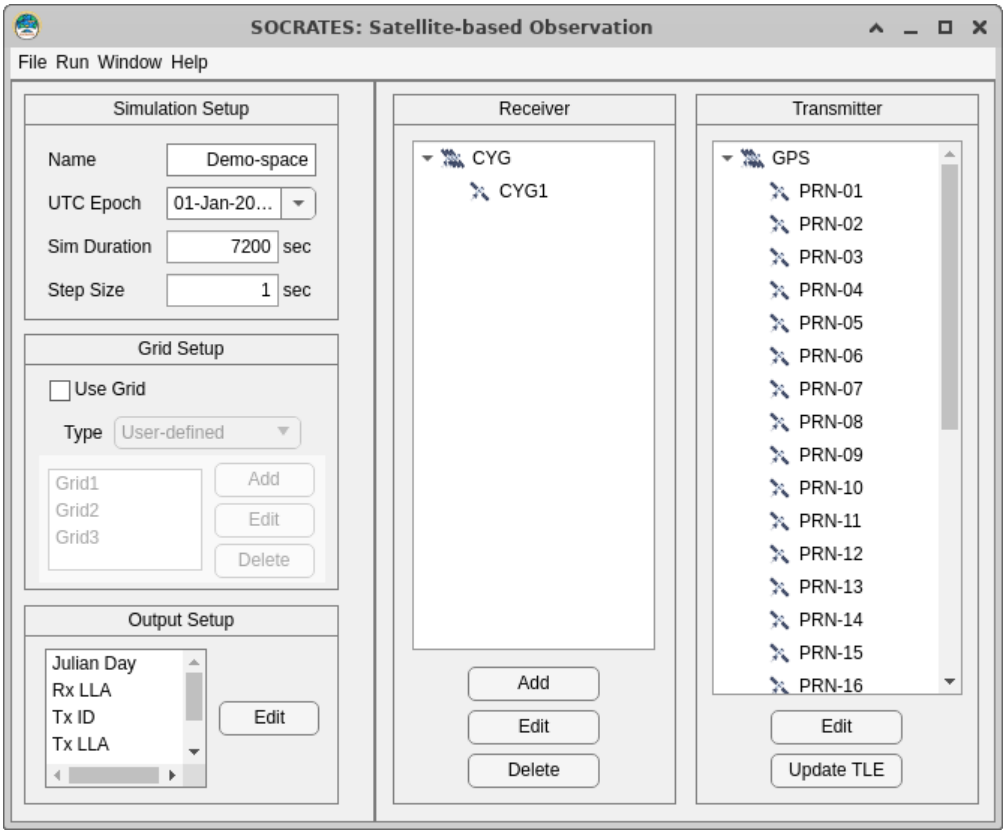



Figure 3.4: Main window for satellite-based observation.

This opens the main window for satellite-based observation as shown in Figure 3.4. By default, the tool loads the SOCRATES config file with the project name (i.e. **Demo-space.smat**) located in the project folder. If this file does not exist, it reads input text files to make the configuration data. The user can open another satellite-based observation project by clicking **File>Open** in the menu bar. The user can load another config file by clicking **File>Load**. It is important to save all changes before closing the window by clicking **File>Save**. This stores all changes to the current config file. The user also can click **File>Save As...** to save the configuration file under a different name. The menu **File>Exit** will close this main window. The following steps will guide the user through each panel in the main window to set up and run the satellite-based observation analysis:

1. **Simulation Setup**

   The user can change the project's name, simulation epoch, duration, and step size. Changing the project name opens a dialog box displaying three options for the user to select: 1) rename the current project, 2) create a new project based on the current project, and 3) cancel. If option 2 is selected, all input folders and files are created for the new project.

2. **Grid Setup**

   The user can place or remove a grid by clicking the "Use Grid" checkbox. One of the two types of grid can be selected from the dropdown menu: Equal-Area Scalable Grid 2.0 (EASE2) and User-defined grid.

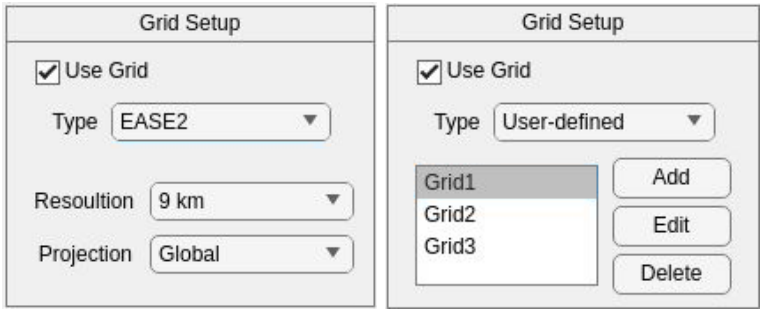



Figure 3.5: Grid setup panel for (left) EASE2 grid and (right) user-defined grid

The EASE2 grid provides various spatial resolutions (1, 3, 3.125, 6.25, 9, 12.5, 25, 36 km) with three projections (Global, Northern Hemisphere, Southern Hemisphere). Refer to [2] for more information about the EASE2 grid.

To set a user-defined grid, click the "Add" button on the Grid Setup panel or select the grid on the list and click the "Edit" button. One or more grids over the regions of interest can be placed. The user-defined grid can be easily defined via the Grid Editor shown in Figure 3.6. The user can change the grid name and the configuration of the grid. Moving and resizing the rectangle on the Grid Map will change the location and size of the grid. Specifying each value on the Grid Editor will also update the grid. Note that an error occurs if the size/location of the grid is invalid.

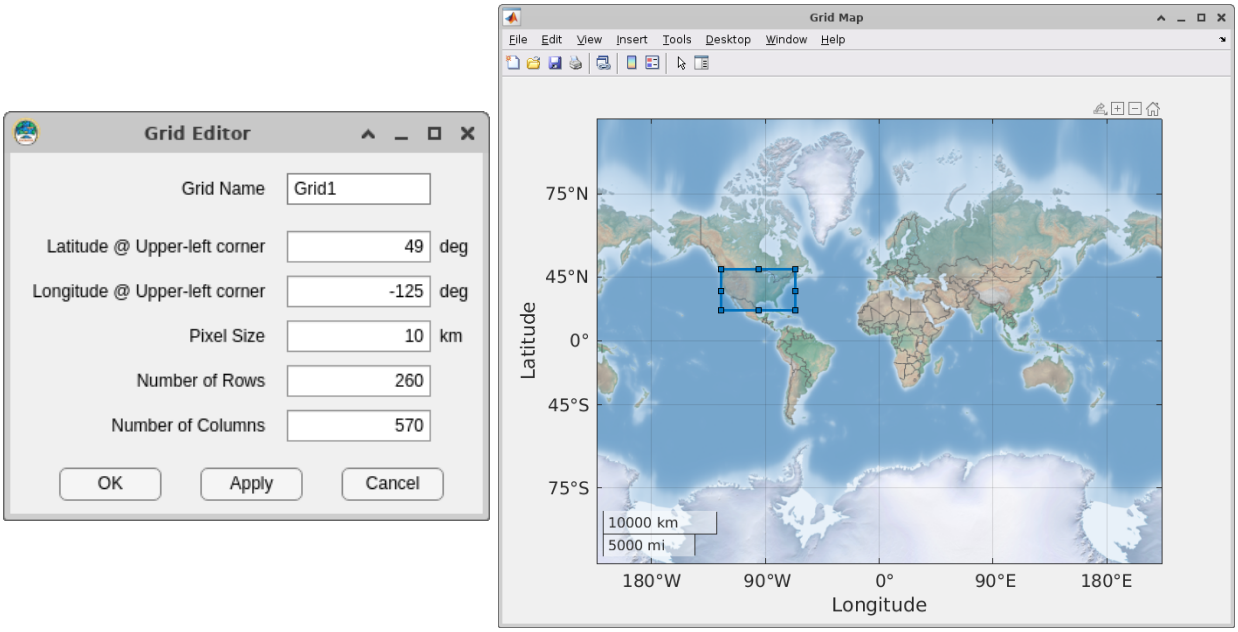



Figure 3.6: Grid Editor

3. **Output Setup**

The user can select output parameters by clicking the "Edit" button in the Output Setup panel. This opens the Report Editor depicted in Figure 3.7. In the General tab, one or multiple output parameters of interest can be selected from the left box and moved to the right box by clicking the ">" button. It can be reversed by the "<" button. In the Subsystems tab, the output for the communication system and attitude determination and control system can be selected. See Section 7.4 and Chapter 8 for more information.

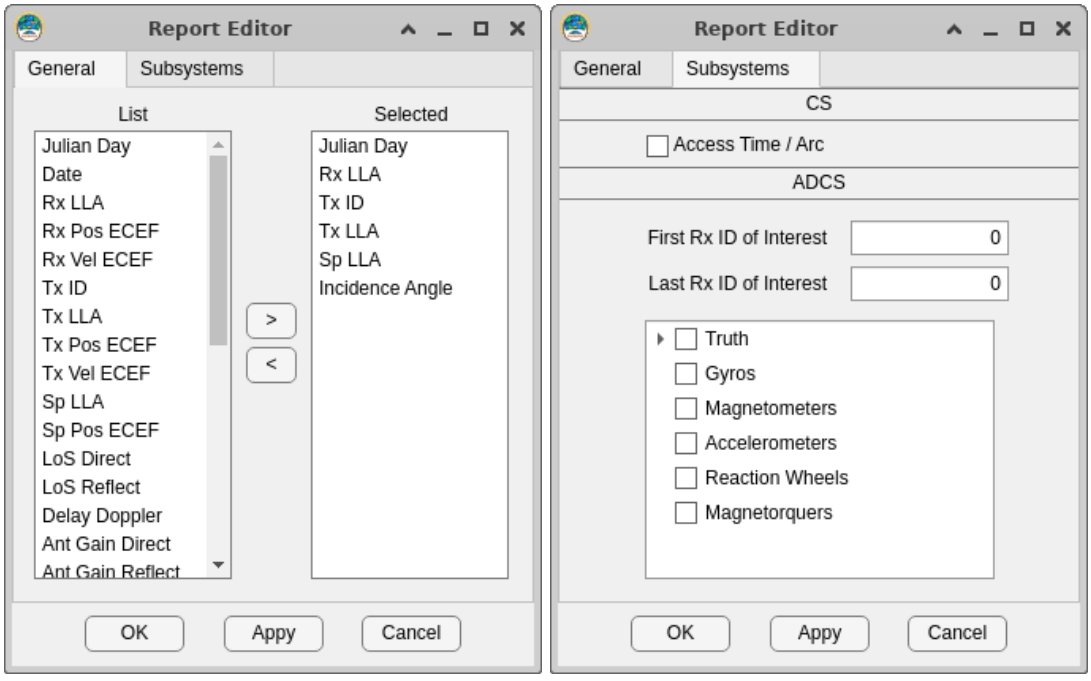



Figure 3.7: Report Editor

4. **Receiver Setup**

   The user can set the number and receivers and their orbits in this Receiver panel. The top-level node represents a constellation. Each child node represents a satellite in that constellation. More satellites can be added by clicking the "Add" button on the Receiver panel. This will open the SC Parameter Editor shown in Figure 3.8. The user can also open the SC Parameters Editor by double-clicking a satellite node or selecting a satellite node and clicking the "Edit" button, or right-clicking on a satellite node to open a context menu and selecting the "Open" menu. A receiver can be deleted by clicking the "Delete" button or selecting the "Delete" menu in the context menu. At least one receiver must exist in a constellation.

   In the SC Parameter Editor, the user can change the receiver's name and select the source of the orbital parameters from the dropdown menu. These sources include Keplerian parameters, position and velocity in the Cartesian coordinates, two-line element set (TLE), and the imported orbit.

   To use TLE, a TLE file named **Inp_TLE_[Receiver Constellation Name].txt** must exist in **./orbit/42input/[project name]** folder. The [**Receiver Constellation Name**] is **CYG** and the [**project name**] is **Demo-space** in this demonstration. A TLE must be labeled with the name of the corresponding satellite node.

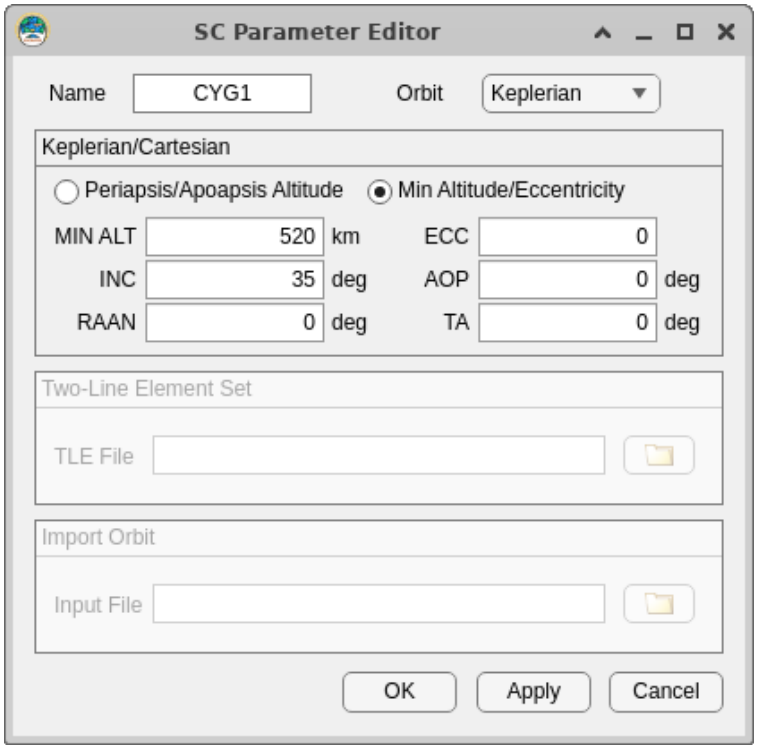



Figure 3.8: SC Parameter Editor.

5. **Transmitter Setup**

   The transmitting constellation can be selected from the existing communication and navigation systems or created by the user. The currently available transmitter systems are the following:

   - GPS
   - Galileo
   - GLONASS
   - Beidou
   - MUOS

- ORBCOMM
- SWARM
- Starlink
- Iridium
- NOAA

On the Transmitter panel, the top-level node represents a constellation and each child node represents a satellite in that constellation. To select the transmitters of interest, click the "Edit" button on the Transmitter panel to open the Transmitter Constellation Editor shown in Figure 3.9.

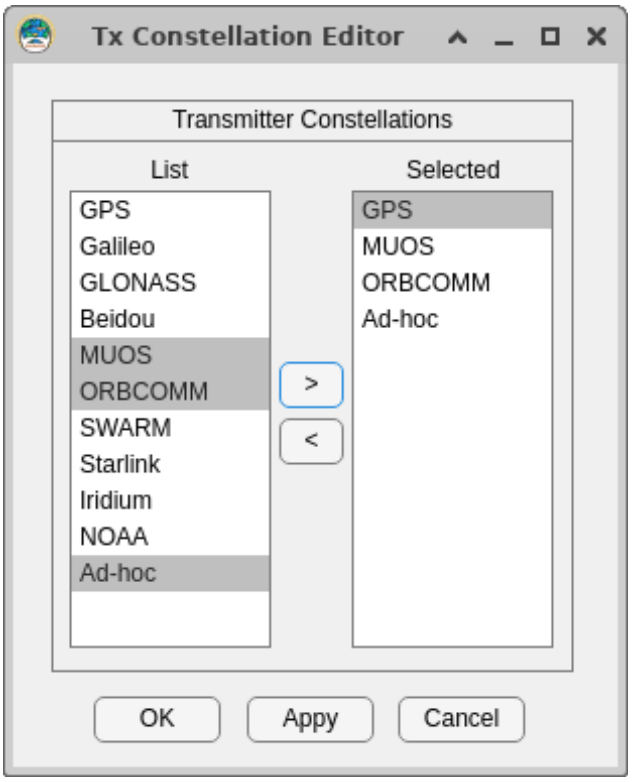



Figure 3.9: Transmitter Constellation Editor.

In this editor, the transmitter constellations listed in the left box can be added to the current transmitter list. For example, the MUOS, ORBCOMM, and Ad-hoc constellations are moved to the right box by clicking the ">" button as shown in Figure 3.9. The user can move one or more nodes left or right. The user can apply the change by clicking the "Apply" button or the "OK" button. The update from this editor will load all satellites listed in the public TLE data from `www.celestrak.com` for each constellation. The required TLE files named **Inp_TLE_[Transmitter Constellation Name].txt** will be created in **./orbit/42input/**[**project name**] folder. The [**project name**] is **Demo-space** in this demonstration. A dialog box will pop up to confirm overwriting any change the user made on the Transmitter panel, such as the addition of satellites and the name change of the nodes.

More satellites can be added by right-clicking on the constellation node and selecting the "Add spacecraft" menu. If the user added an ad-hoc constellation, he/she must add a satellite node and delete the message node created with the constellation node. To add a satellite to the ad-hoc constellation, right-click on the constellation node and select the "Add spacecraft" menu, as shown in Figure 3.10.

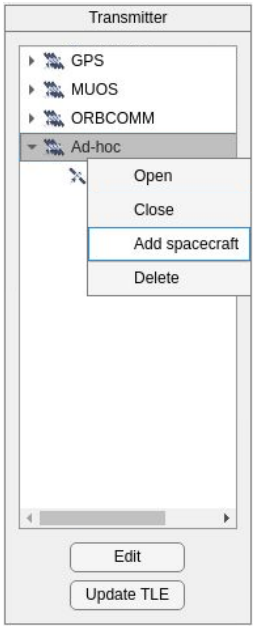



Figure 3.10: Adding a satellite to Ad-hoc constellation.

The SC Parameter Editor will automatically pop up for user input. To delete a constellation or a satellite, right-click on the corresponding node and select the "Delete" menu. Note that the user can update the TLE input files with the latest public data by clicking the "Update TLE" button on the Transmitter panel.

6. **Instrument Setup**

   The properties of receivers and transmitters must be set for each transmitting constellation. This can be done via the SoOp Parameter Editor. To open the SoOp Parameter Editor, right-click on a transmitter constellation node on the Transmitter panel to open a context menu. Then select the "Open" menu as shown in Figure 3.11. The user can also open it by double-clicking the transmitter constellation node. The "Close" menu will close the opened SoOp Parameter Editor. The SoOp Parameter Editor defines the following properties for each transmitter-receiver pair:

   - Transmitter: frequency, polarization, Effective Isotropic Radiated Power (EIRP), bandwidth.
   - Receiver: Maximum number of specular points simultaneously trackable, maximum antenna gain, polarization, antenna radiation pattern.

   The antenna pattern for the receiver can be selected from three options: Ideal, Generalized-Gaussian, and User-defined.

   For the Generalized-Gaussian antenna pattern, the following inputs must be provided:

   - HPBW: Half-power beamwidth of the antenna pattern in degrees.
   - Resolution: The minimum sensitivity of the antenna pattern in degrees.
   - Sidelobe level: The level of the first side-lobe in decibels.
   - Cross-pol level: The level of the cross-polarization of the antenna pattern in decibels.

   For the User-defined antenna pattern, antenna pattern files are required. Example antenna pattern files are located in **./ant_pat** folder. Two .csv files are required, each with normalized voltage values for co-polarization and cross-polarization. The antenna pattern should be provided such that columns represent the $\theta$-angle look-up and rows represent the $\phi$-angle look-up. The $\theta$ angles have a 180° scan, while the $\phi$ angles have a 360° scan.

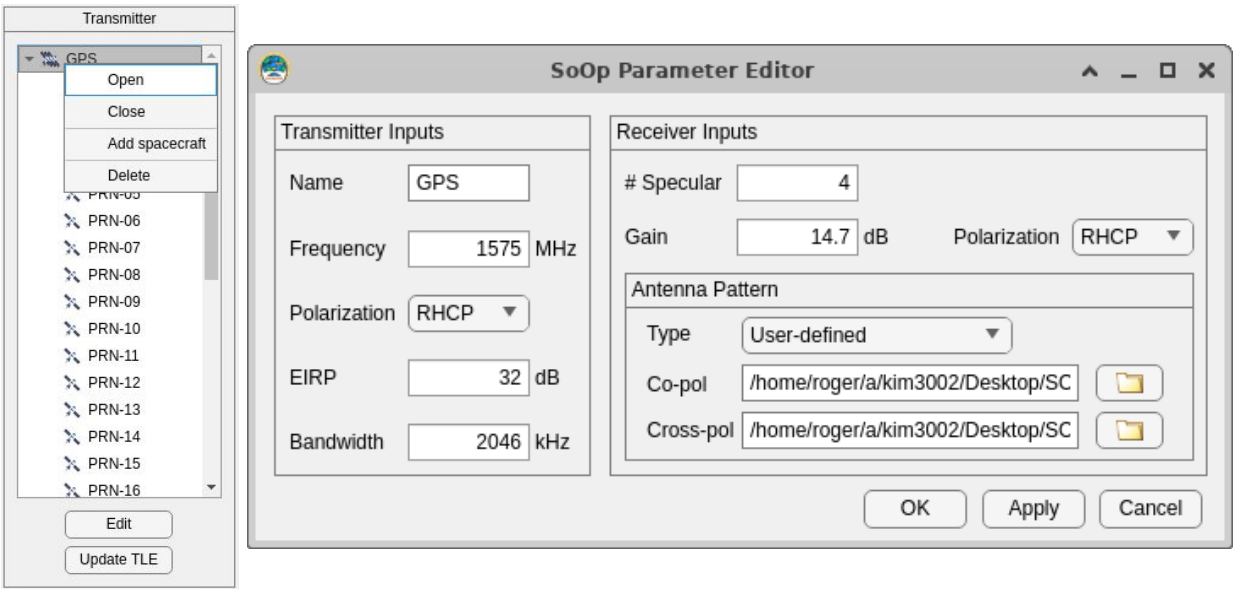



Figure 3.11: Opening the SoOp Parameter Editor.

7. **Visualization**

The 3-D visualization of satellite constellations is available. This requires MATLAB Satellite Communications Toolbox. To visualize the constellations, click **Run>Visualization**.

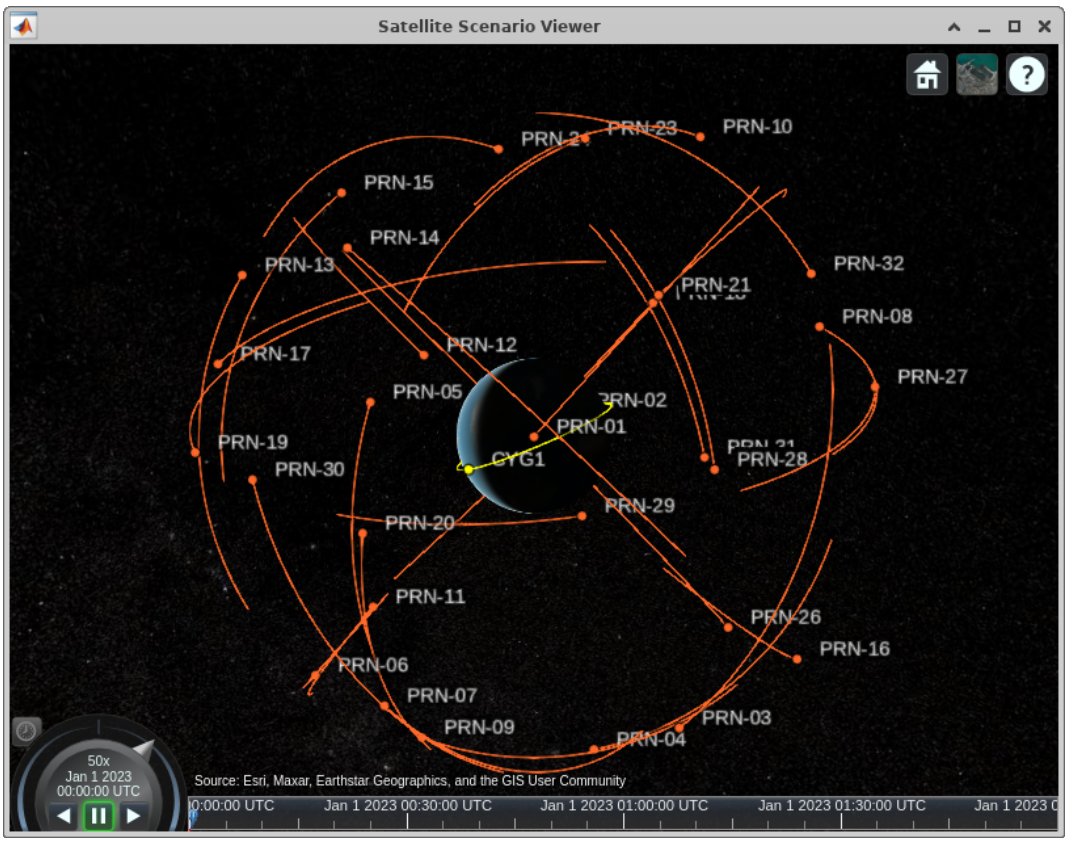



Figure 3.12: Visualization of satellite constellations.

8. **Compilation**

   The user can compile the SOCRATES by clicking **Run>Compile**.

9. **Generation of Input Files**

   The user can generate necessary input files for the SOCRATES by clicking **Run>Generate input files**. A dialog box will pop up to confirm overwriting the existing input files. These input files include **Inp_Main.txt**, **Inp_Inorbit.txt**, **Inp_Report.txt**, and **Inp_Sim.txt**. The first three files are located in the project folder, (i.e. **./sim/Demo-space**). The last file is located in the 42 orbit folder, (i.e. **./orbit/42input/Demo-space**). If the user has made any change in any panel, the input files must be re-generated. To proceed with the generation, click the "Yes" button. To cancel, click the "No" button.

   If the user clicked "Yes", the following dialog box asks whether the SC and Orb input files for the 42 simulator should also be generated. This will update any change in the name and orbital parameters of the satellites. The user can create files only for receivers, only files for transmitters, or create or cancel both.

10. **Run**

    The user can run the simulation by clicking **Run>Run**. The simulation progress is displayed in MATLAB's Command Window. The output files will be located in the **./sim/Demo-space/out** folder.

# Chapter 4

# Tower-based Observation

The provided simulation project for the tower-based observation is multi-frequency SoOp-R. A SoOp-R receiver is installed on top of an 8.72-meter tower located at the Purdue Agronomy Center for Research and Education (ACRE) at 40.474398°N, 86.991735°W. The transmitting systems of interest are commercial Orbcomm satellites for I-band transmission and US Navy's Mobile User Objective System (MUOS) for P-band transmissions. The SOCRATES will determine the specular reflection points and generate outputs that are requested in **Inp_Report.txt**. The simulation duration is 6 hours.

## 4.1 Run in Terminal

1. The included input files are configured for the tower-based observation by default. If the user wants to see and modify output parameters, open the **Inp_Report.txt** located in **./sim/Demo-tower** folder and select output parameters of interest. `TRUE`/`FALSE` turns on/off the corresponding parameter.

2. Open the command prompt at the top-level directory and enter the command:

   ```
   $ SOCRATES Demo-tower
   ```

   It will load the simulation inputs located in the **./sim/Demo-tower** folder.

```
kim3002@cadia:~/Desktop/SOCRATES$ SOCRATES Demo-tower
Attempted divide by zero in UNITV (Line 337 of mathkit.c)
Attempted divide by zero in UNITV (Line 337 of mathkit.c)
Reached CmdScript EOF at Time = 0.000000
/*********************************************************************/
/*                                                                   */
/*                              SOCRATES                             */
/*                                                                   */
/*  SoOp Constellation and Remote sensing Analysis Tool for Earth Science  */
/*                                                                   */
/*********************************************************************/
>> Number of Input Soil Layers is 1
>> Initiate In-Orbit Observation ...
>> - Number of Receivers : 1
>> - Number of Transmitter Systems : 2
>> - Orbit Propagator : 42
    42 Case ./orbit/42input/Demo-tower/ is   0% Complete at Time =        0.000
    42 Case ./orbit/42input/Demo-tower/ is  10% Complete at Time =     2160.000
    42 Case ./orbit/42input/Demo-tower/ is  20% Complete at Time =     4320.000
    42 Case ./orbit/42input/Demo-tower/ is  30% Complete at Time =     6480.000
    42 Case ./orbit/42input/Demo-tower/ is  40% Complete at Time =     8640.000
    42 Case ./orbit/42input/Demo-tower/ is  50% Complete at Time =    10800.000
    42 Case ./orbit/42input/Demo-tower/ is  60% Complete at Time =    12960.000
    42 Case ./orbit/42input/Demo-tower/ is  70% Complete at Time =    15120.000
    42 Case ./orbit/42input/Demo-tower/ is  80% Complete at Time =    17280.000
    42 Case ./orbit/42input/Demo-tower/ is  90% Complete at Time =    19440.000
    42 Case ./orbit/42input/Demo-tower/ is 100% Complete at Time =    21600.000
     Total Run Time =    327.00 sec
     Sim Speed =    66.06 x Real
>> Simulation End.
kim3002@cadia:~/Desktop/SOCRATES$
```

Figure 4.1: Execution screenshot of tower-based observation demo.

3. After completion, the output files will be located in the **./sim/Demo-tower/out** folder. Figure 4.2 shows the example of output files. See Section 7.4 for the naming convention and description of the output file. Note that the outputs related to the specular points including the position and angles have `nan` values if the elevation angle of the corresponding transmitter is less than 1°.

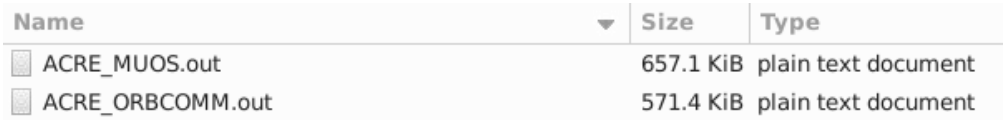



Figure 4.2: Products of tower-based observation.

## 4.2 Run Using GUI

This feature is not supported yet.

# Chapter 5

# Fixed Observation

The included simulation project for the fixed observation is P-band (255 MHz and 370 MHz) observation with a receiver at 400 km and a geostationary transmitter. The polarization of the receiver's antenna is dual circular polarization and the polarization of the transmitter's antenna is right-handed circular polarization (RHCP). The incidence angle is fixed at 30°. The orientation of the receiver is facing in the direction of specular reflection. The simulator will generate values for the received direct signal's power density and electric field and will also generate reflectivity values for the received coherently scattered field in both co-polarization (RHCP transmit and RHCP receive) and cross-polarization (RHCP transmit and LHCP receive).

## 5.1 Run in Terminal

1. The included input files are configured for the fixed observation by default.
2. Open the command prompt at the top-level directory and enter the command:

   ```
   $ SOCRATES Demo-fixed
   ```

   It will load the simulation inputs located in the **./sim/Demo-fixed** folder.

```
kim3002@cadia:~/Desktop/SOCRATES$ SOCRATES Demo-fixed
/*************************************************************************/
/*                                                                       */
/*                               SOCRATES                                */
/*                                                                       */
/*  SoOp Constellation and Remote sensing Analysis Tool for Earth Science  */
/*                                                                       */
/*************************************************************************/
>> Number of Input Soil Layers is 1
>> Initiate Fixed Geometry Observation ...
>> - Total # of Data = 60

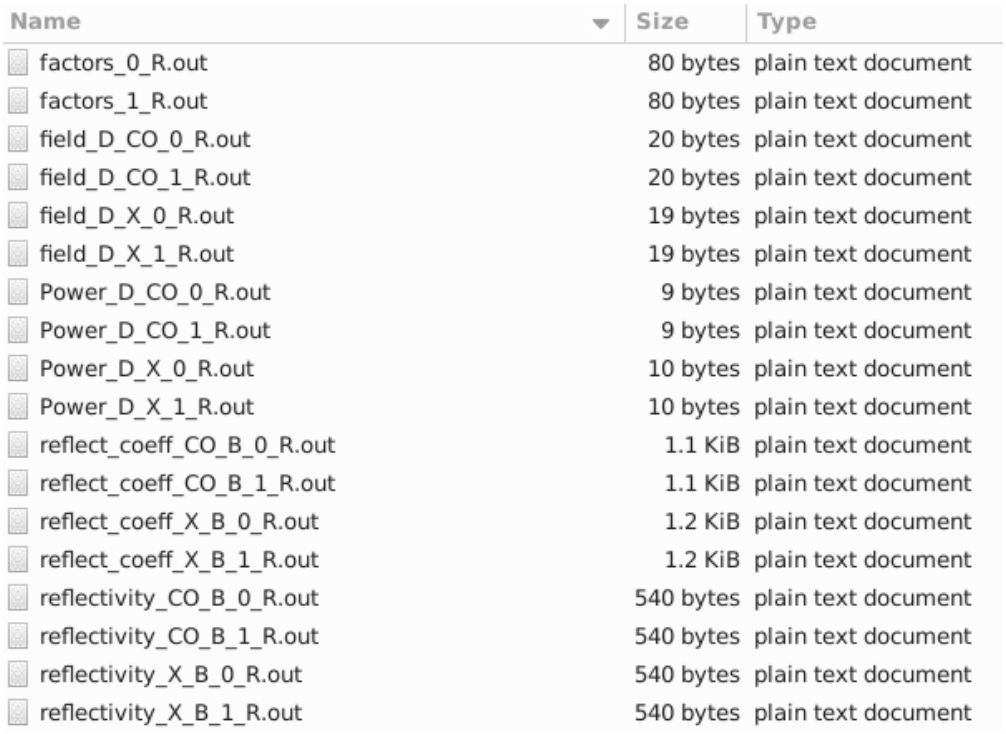
>> START SIMULATIONS

>> SoOp Configuration #1: Tx Freq = 255.000000 MHz, Rx Pol = R
>> SoOp ./sim/Demo-fixed/ is   0% Complete
>> SoOp ./sim/Demo-fixed/ is  10% Complete
>> SoOp ./sim/Demo-fixed/ is  20% Complete
>> SoOp ./sim/Demo-fixed/ is  30% Complete
>> SoOp ./sim/Demo-fixed/ is  40% Complete
>> SoOp ./sim/Demo-fixed/ is  50% Complete
>> SoOp ./sim/Demo-fixed/ is  60% Complete
>> SoOp ./sim/Demo-fixed/ is  70% Complete
>> SoOp ./sim/Demo-fixed/ is  80% Complete
>> SoOp ./sim/Demo-fixed/ is  90% Complete
>> SoOp ./sim/Demo-fixed/ is 100% Complete

>> SoOp Configuration #2: Tx Freq = 370.000000 MHz, Rx Pol = R
>> SoOp ./sim/Demo-fixed/ is   0% Complete
>> SoOp ./sim/Demo-fixed/ is  10% Complete
>> SoOp ./sim/Demo-fixed/ is  20% Complete
>> SoOp ./sim/Demo-fixed/ is  30% Complete
>> SoOp ./sim/Demo-fixed/ is  40% Complete
>> SoOp ./sim/Demo-fixed/ is  50% Complete
>> SoOp ./sim/Demo-fixed/ is  60% Complete
>> SoOp ./sim/Demo-fixed/ is  70% Complete
>> SoOp ./sim/Demo-fixed/ is  80% Complete
>> SoOp ./sim/Demo-fixed/ is  90% Complete
>> SoOp ./sim/Demo-fixed/ is 100% Complete
>> Simulation End.
kim3002@cadia:~/Desktop/SOCRATES$
```

Figure 5.1: Execution screenshot of fixed observation demo.

3. After completion, the output files will be located in the **./sim/Demo-fixed/out** folder:



| Name | Size | Type |
|---|---|---|
| factors_0_R.out | 80 bytes | plain text document |
| factors_1_R.out | 80 bytes | plain text document |
| field_D_CO_0_R.out | 20 bytes | plain text document |
| field_D_CO_1_R.out | 20 bytes | plain text document |
| field_D_X_0_R.out | 19 bytes | plain text document |
| field_D_X_1_R.out | 19 bytes | plain text document |
| Power_D_CO_0_R.out | 9 bytes | plain text document |
| Power_D_CO_1_R.out | 9 bytes | plain text document |
| Power_D_X_0_R.out | 10 bytes | plain text document |
| Power_D_X_1_R.out | 10 bytes | plain text document |
| reflect_coeff_CO_B_0_R.out | 1.1 KiB | plain text document |
| reflect_coeff_CO_B_1_R.out | 1.1 KiB | plain text document |
| reflect_coeff_X_B_0_R.out | 1.2 KiB | plain text document |
| reflect_coeff_X_B_1_R.out | 1.2 KiB | plain text document |
| reflectivity_CO_B_0_R.out | 540 bytes | plain text document |
| reflectivity_CO_B_1_R.out | 540 bytes | plain text document |
| reflectivity_X_B_0_R.out | 540 bytes | plain text document |
| reflectivity_X_B_1_R.out | 540 bytes | plain text document |

Figure 5.2: Products of fixed observation.

- **field_D_CO.out**: The real and imaginary parts of the direct electric field for co-polarization seen by the receiver.
- **field_D_X.out**: The real and imaginary parts of the direct electric field for cross-polarization seen by the receiver.
- **Power_D_CO.out**: The power of the direct signal for co-polarization received by the receiver.
- **Power_D_X.out**: The power of the direct signal for cross-polarization received by the receiver.
- **reflect_coeff_CO.out**: The real and imaginary parts of reflection coefficients for co-polarization seen by the receiver.
- **reflect_coeff_CO.out**: The real and imaginary parts of reflection coefficients for cross-polarization seen by the receiver.
- **reflectivity_CO.out**: The reflectivity for co-polarization seen by the receiver.
- **reflectivity_X.out**: The reflectivity for cross-polarization seen by the receiver.
- **factors.out**: The real and imaginary parts of "K", "Kd", and "Kc" constants (described in detail in [3]).

The suffix **0** and **1** indicate the SoOp configuration number, where **0** corresponds to 255 MHz and **1** corresponds to 370 MHz. The suffix **R** denotes the polarization of the receiver's antenna. The suffix **B** represents the bare soil environment.

## 5.2 Run Using GUI

This feature is not supported yet.

# Chapter 6

# Mission Design

In this Chapter, we introduce the mission design tool for multi-SoOp constellations. This tool supports the exploration of tradespace and the high-level mission design of multi-SoOp constellations. The overview of the framework is presented in Figure 6.1. Major steps include *specular point prediction*, *figure of merit calculation*, and *score mapping*. Note that the SOCRATES GUI is required to complete the entire design process. The SOCRATES GUI will guide the user through each of the processes and provide an easy setup and analysis environment. A detailed step-by-step guide for running the demo project is provided in the following sections.

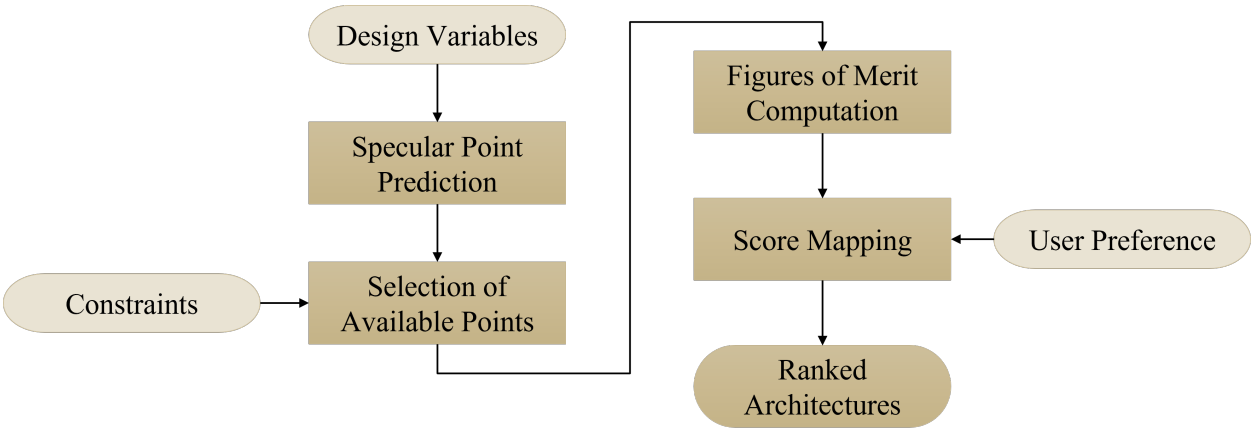



Figure 6.1: Framework for multi-SoOp constellation design.

## 6.1 Opening the Mission Design Tool

To open the mission design tool, the user first needs to run **SOCRATES.mlapp** located in the top-level directory. The current folder of MATLAB must be set to the top-level directory of the SOCRATES (i.e. **./**). From the SOCRATES main window, click the "Mission Design" button to select the mission design tool. This will open a path selector as shown in Figure 6.2. Select **Demo-design** folder to open the demo project and click the "Open" button. The user must select a project configured for mission design. This is set in **Inp_Main.txt**.

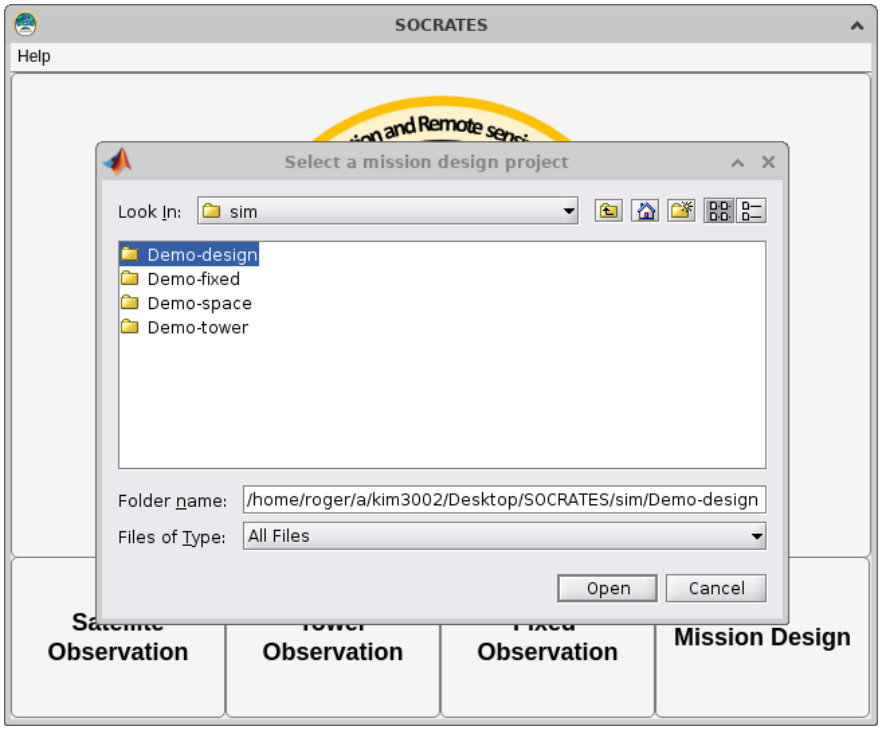



Figure 6.2: Project selection window for the mission design tool.

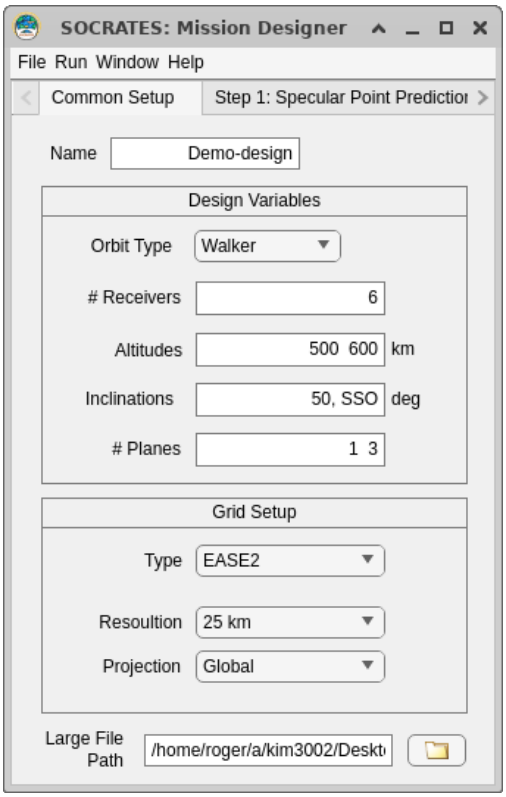



Figure 6.3: Main window for the mission design tool.

This opens the main window of the mission design tool as shown in Figure 6.3. By default, the tool loads the SOCRATES config file with the project name (i.e. **Demo-design.smat**) located in the project folder. If this file does not exist, it reads input text files to make the configuration data. The user can open another mission design project by clicking **File>Open** in the menu bar. The user can load another config file by clicking **File>Load**. It is important to save all changes before closing the window by clicking **File>Save**. This stores all changes to the current config file. The user also can click **File>Save As...** to save the configuration file under a different name. The menu **File>Exit** will close this main window.

## 6.2 Common Setup

The main window of the mission design tool provides the input panels for a common setup. It is where all the common configurations for the project's name, design variables, and grid cells are defined. In the Common Setup tab, the user can configure the followings:

1. **Project's Name**

   The user can change the project's name by simply re-typing the name. Changing the project name opens a dialog box displaying three options for the user to select: 1) rename the current project, 2) create a new project based on the current project, and 3) cancel. If option 2 is selected, all input folders and files are created for the new project.

2. **Design Variables**

   The current version of the mission design tool supports the Walker constellation and the hybrid constellation for receiving satellites. Figure 6.4 shows the design variable panel.

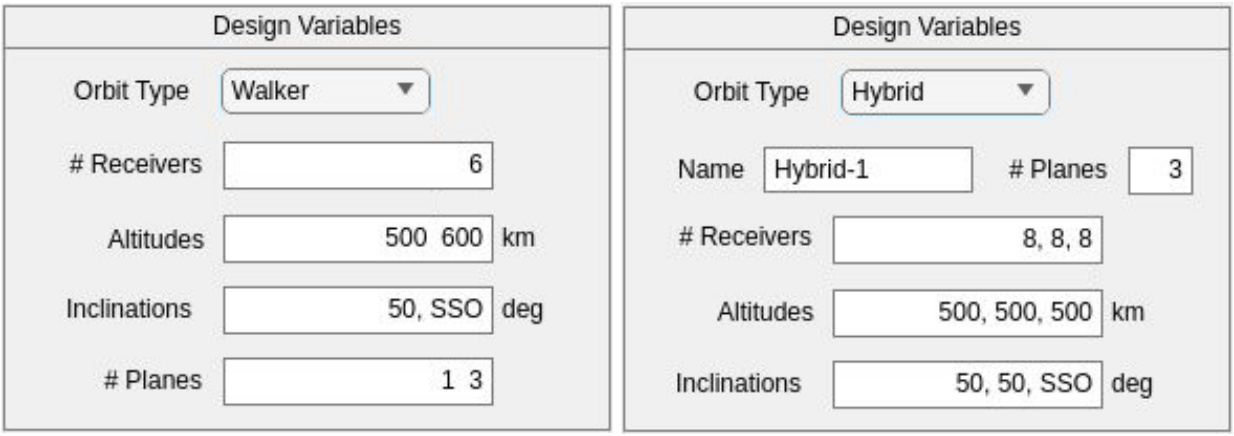



Figure 6.4: Design variable setup panel for (left) Walker constellation and (right) hybrid constellation.

For the Walker constellation, the user must set the name of the constellation and the range of the following design variables:

- Total number of satellites;
- Altitude of satellites;
- Inclination of orbital planes;
- Number of equally spaced orbital planes.

Each value should be separated by a comma. All combinations of each value of the design variables will be simulated as design alternatives. The naming convention of a Walker constellation is [**# Satellites**]_[**Altitude**]_[**Inclination**]_[**# Planes**]. The name of the Walker constellation will be used in the output directories and files in the specular point prediction step and the figure of merit calculation step.

For the hybrid constellation, the user must define the number of orbital planes and the following parameters for each orbital plane:

- Number of satellites in each orbital plane;
- Altitude of satellites in each orbital plane;
- Inclination of each orbital plane.

Note that the tool can simulate only one hybrid constellation at a time. The name of a hybrid constellation defined by the user will be used in the output directories and files in the specular point prediction step and the figure of merit calculation step.

3. **Grid Setup**

   For coverage analysis, a grid must be set in the mission design. The grid can be selected as either the EASE Grid 2.0 or user-defined grids. For detailed information, please refer to the Grid Setup item in Section 3.2.

4. **Large File Path**

   The Large File Path is a directory for writing and reading all output files in the mission design. The user should select a directory with sufficient memory or disk space.

## 6.3 Specular point prediction

This step selects the transmitting systems of interest and propagates the specular reflection points. It is fundamentally the same process as the satellite-based observation. But the receiving constellation is automatically generated based on the selected design variables. This step supports the MPI-based parallel processing of the propagation of multiple constellations.

The Specular Point Prediction tab is shown in Figure 6.5. Follow each of the steps below to run the specular point prediction simulation:

1. **Configuring Simulation Settings**

   The user can configure the simulation epoch, duration, step size, and the number of parallel processes. If single processing is used, the number of processes must be set to 1. If multi-processing is used, multiple processors simultaneously propagate design alternatives assigned to each processor. One processor can process one design alternative at a time. As an option, the user can generate the report output file for each receiver by checking on the "Report Rx data" checkbox. The report data can be selected in the Report Editor by clicking the "Edit Report" button. Note that generating these report files may require a large amount of memory for storage. Use only when necessary. Please refer to the Output Setup in Section 3.2 for detailed information on the Report Editor.

2. **Configuring Transmitter Settings**

   The user can select the transmitter systems of interest and update the TLE data. Checking on the "Use existing trajectories" checkbox reuse the transmitter position data generated from the previous specular point prediction simulation. It is **strongly recommended** that the user run a single-processed simulation with a single receiving constellation (it can be one dummy receiver) to generate the trajectories of the transmitting satellites. Then the user can add all the design alternatives and check on the "User existing trajectories" checkbox to reuse the transmitters' position. The transmitter position data will be stored in **./sim/Demo-design/out/TxOrb** folder. In that way, the user can avoid wasting the computational power of repeatedly propagating the trajectories of the transmitters. Please refer to the Transmitter Setup and Instrument Setup in Section 3.2 for detailed information on the Transmitter Constellation Editor and the SoOp Parameter Editor.

3. **Running Speuclar Point Simulation**

   The user can run the specular point prediction simulation by clicking the "Run" button in the Specular Point Prediction tab. This will prompt **File>Save** and **Run>Generate input files** automatically.

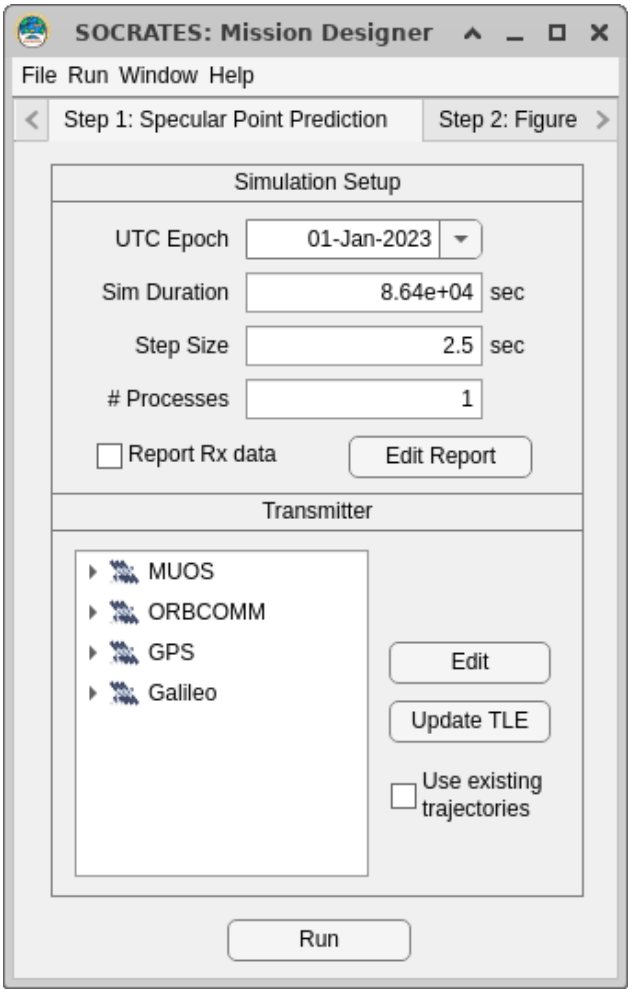



Figure 6.5: Specular Point Prediction tab of the mission design tool.

It will store all changes to the current config file and open up a dialog box to confirm overwriting the existing input files before running the simulation. Please refer to the Generation of Input Files in Section 3.2 for detailed information.

The user can also run this by opening a terminal on the top-level directory and entering the command:

```
$ mpirun -np [# Processes] SOCRATES Demo-design prop
```

### 6.3.1 Output

The output files of this step are the netCDF files containing the following data:

- Access time;
- Latitude/longitude of the specular point;
- Azimuth/elevation angles of the incoming direct/reflected signal;
- Range from the transmitter to the receiver;
- Range from the transmitter to the specular point;
- Range from the specular point to the receiver.

The output file is generated for each receiver in the location of the Large File Path defined in the Common Setup tab. The naming convention of a subdirectory and an output file is

**[Large File Path]/[Constellation Name]/[Constellation Name]_Rx[Satellite Number].nc**

This file is a prerequisite to computing the intermediate variables in the next step.

## 6.4 Figure of Merit Calculation

This step sets the constraints for selecting valid specular points, selects the transmitter combinations and frequency combinations, and computes intermediate variables and figures of merit (FoMs). Figure 6.6 shows the Figure of Merit tab.

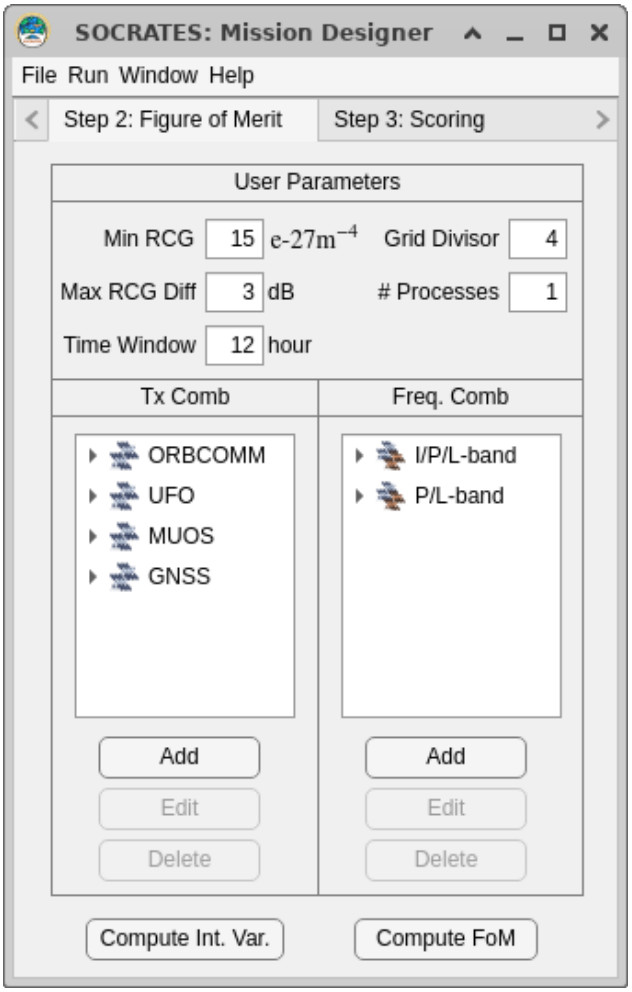



Figure 6.6: Figure of Merit tab of the mission design tool.

Follow each of the steps below to compute the FoMs:

1. **Configuring User Parameter Settings**

   In the User Parameters panel, the user should set the constraints for each transmitter combination to filter out any invalid specular points where the signal strength is too weak to retrieve accurate observation. For GNSS-R, the user should set the minimum RCG. For SoOp, the user should set the maximum RCG difference between the direct signal and the reflected signal. For multi-frequency

observation, the user should set the observation time window. The grid divisor is intended to calculate coverage by dividing the EASE grid into multiple grids of the same size to reduce memory usage. If the user utilizes a user-defined grid, the grid divisor must be 1. The computation of the intermediate variables can be multi-processed. If single processing is used, the number of processes must be set to 1. If multi-processing is used, multiple processors simultaneously compute the intermediate variables of design alternatives assigned to each processor. One processor can process one design alternative at a time.

2. **Configuring Transmitter Combination Settings**

   A *transmitter combination* refers to a group of transmitters using the same frequency. For example, the user may want to group the GPS and Galileo as one group of transmitters using L-band signals. In this case, the user can click the "Add" button in the Tx Comb panel to open the Transmitter Combination Editor shown in Figure 6.7. Name the transmitter combination as "GNSS" and set the receiver inputs. Then select the GPS and Galileo on the left box and move them to the right box to group them as a transmitter combination. Click the "OK" button to save the transmitter combination. The transmitter combination can also define a case where a satellite transmits signals at different frequencies. In this case, because each frequency uses a different antenna, a transmitter combination must be set for each frequency. For example, the MUOS satellite operates multiple narrow-band channels in a legacy UHF Follow-On (UFO) band at 240–270 MHz and four wideband code division multiple access (WCDMA) channels between 360–380 MHz. Then the user can add two transmitter combinations both with the MUOS constellation but one for UFO channels and the other for WCDMA channels. The user can also edit and delete any transmitter combination by selecting its node and clicking the "Edit" button and the "Delete" button in the Tx Comb panel.

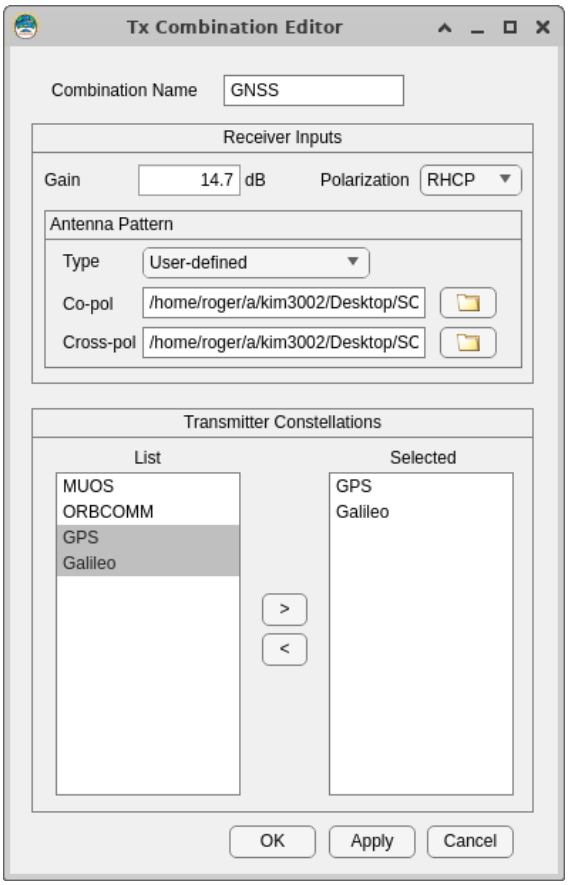



Figure 6.7: Transmitter Combination Editor.

3. **Configuring Frequency Combination Settings**

   A *frequency combination* refers to a group of transmitter combinations that are required to establish a multi-frequency observation. The user can define one or more frequency combinations to analyze the coverage of multi-frequency observations with various combinations. For example, the user can add the P/L-band observation by clicking the "Add" button in the Freq. Comb panel to open the Frequency Combination Editor shown in Figure 6.8. Name the frequency combination as "P/L-band" and move the UFO, MUOS, and GNSS to the right box to group them as a frequency combination. Then click the "OK" button to save it. The user can also edit and delete any frequency combination by selecting its node and clicking the "Edit" button and the "Delete" button in the Freq. Comb panel.

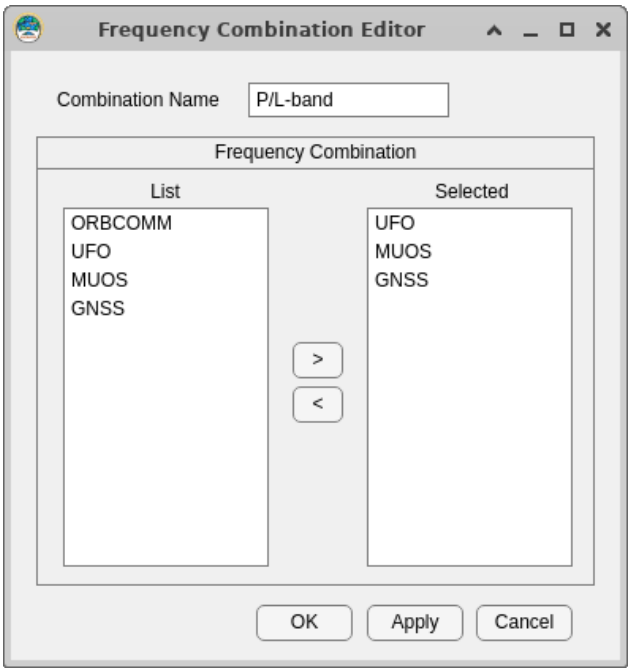



Figure 6.8: Frequency Combination Editor.

4. **Computing Intermediate Variables**

   A prerequisite to computing the figures of merit (FoMs) is to compute intermediate variables for each transmitting constellation using the output files generated from the specular point prediction step. Click the "Compute Int. Var." button in the Figure of Merit tab to generate NetCDF files containing the intermediate variables. The user can also run this by opening a terminal on the top-level directory and entering the command:

   ```
   $ mpirun -np [# Processes] SOCRATES Demo-design fom
   ```

   The cost file and the performance file can be generated separately by entering the command:

   ```
   $ mpirun -np [# Processes] SOCRATES Demo-design cost
   ```

   for the cost file and

   ```
   $ mpirun -np [# Processes] SOCRATES Demo-design perf
   ```

   for the performance file.

5. **Computing FoMs**

Once the intermediate variables are computed, click the "Compute FoM" button in the Figure of Merit tab to compute the FoMs. A dialog box will appear to confirm overwriting the FoM output file. Click the "OK" button to proceed. Otherwise, click the "Cancel" button.

### 6.4.1 Output

**Intermediate Variables**

The intermediate variables for performance include:

- 1st access time for each grid cell;
- 2nd access time for each grid cell;
- Revisit times for each grid cell.

The intermediate variables for cost include:

- Constellation cost;
- Dedicated launch cost;
- Rideshare launch cost;
- Launch option;
- Number of launches.

The intermediate variables are stored in two separate NetCDF files for performance and cost:

**[Large File Path]/[Constellation Name]/[Constellation Name]_perform.nc**

**[Large File Path]/[Constellation Name]/[Constellation Name]_cost.nc**

These files are prerequisites to computing the FoMs.

**Figures of Merit**

The top-level FoM categories are cost and performance. The cost category provides three metrics:

- Total Cost: This metric is the sum of the constellation cost and the launch cost.
- Constellation Cost: This metric is the cost estimate of building satellites in a constellation. It accounts for productivity improvement.
- Launch Cost: This metric is the cost estimate of launching satellites in a constellation. It accounts for the maximum mass per launch. The Rocket Lab's Electron launch vehicle is used as a reference launch vehicle.

Note that the cost model used in this tool provides relative cost estimates between different constellations, rather than accurate absolute cost estimates. The user can select one of these three metrics for the cost evaluation.

The performance category comprises several subcategories that evaluate different aspects of the system's performance:

- Multi-frequency spatial coverage (MSC): This metric measures the percentage of grid cells that have at least one multi-frequency observation. A multi-frequency observation is achieved when specular points of all frequencies of interest are located on a grid cell within a given observation time window. In the demo project, we consider the combination of all four frequencies (denoted as "I/P/L-band") and the combination of three frequencies in the P/L-band (denoted as "P/L-band"). The observation time window was set to 12 hours.

- Multi-frequency temporal coverage (MTC): This metric calculates the average number of observation time windows that contain at least one multi-frequency observation over all grid cells. If a grid cell has one or more multi-frequency observations at a given observation time window, it is noted as one (1), otherwise a zero (0). This value is accumulated over the simulation period for each grid cell, and the MTC is then calculated by averaging these values of all grid cells. Given the observation time window and simulation time used in the demo project (12 hours and 1 day, respectively), the maximum MTC is 2.
- Max coverage: This metric computes the percentage of grid cells that have at least one specular point at the end time of the simulation. It is calculated for each transmitting constellation.
- Aggregate revisit time: This metric assesses the temporal gap between two successive specular points of a transmitting constellation at a grid cell. It is calculated for each transmitting constellation. The available aggregate revisit times are mean, median, and high percentile (90th/95th/98th) revisit times. Note that a revisit time shorter than 1 hour is not considered in the aggregation.

The FoM metrics are stored in a MAT file:

**./sim/Demo-design/out/FoM_Demo-design.mat**

This file will be used to compute scores in the next step.

## 6.5 Score Mapping

This step converts the FoMs to normalized scores ranging from 0 to 1, a single numerical measure to compare the design alternatives. Figure 6.9 illustrates the score mapping process for each design alternative. The first part of the diagram represents a hierarchical FoM structure. Each FoM is converted to a score based on each scoring function. Wymore's standard scoring functions are used in score mapping. The "More is better" type scoring function (denoted as SSF3) is used for MSC, MTC, and max coverage. The "More is worse" type scoring function (denoted as SSF9) is used for cost and revisit time. Then each FoM is weighted and summed up for each layer. This yields the final score for each design candidate.

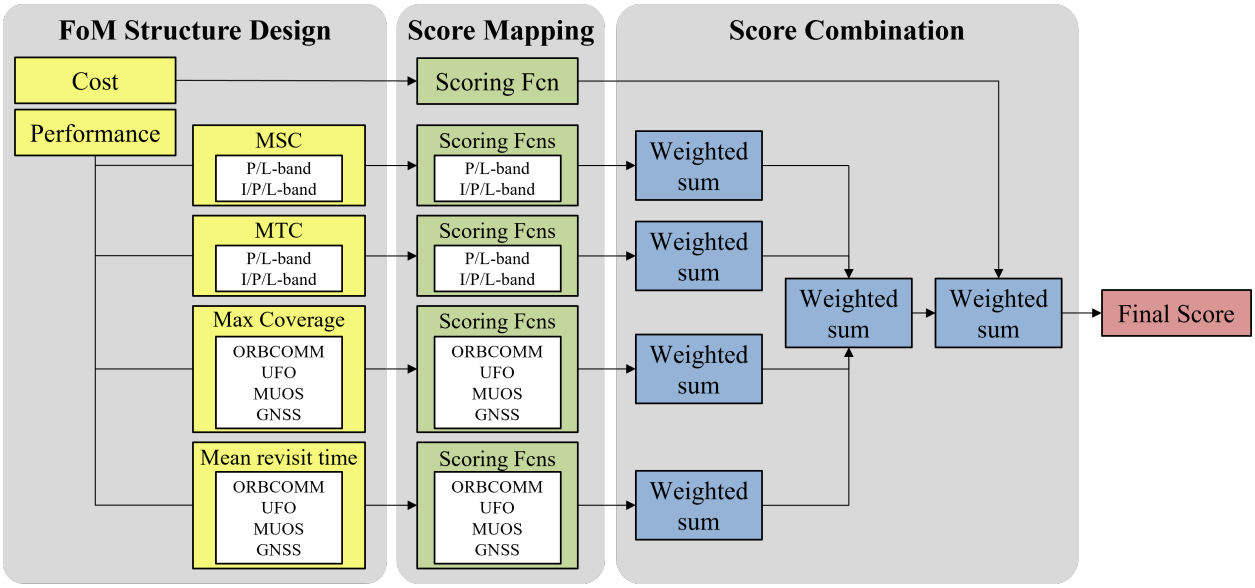



Figure 6.9: A diagram for score mapping of a design alternative.

The weights and parameters for scoring functions can be set in the Scoring tab shown in Figure 6.10.

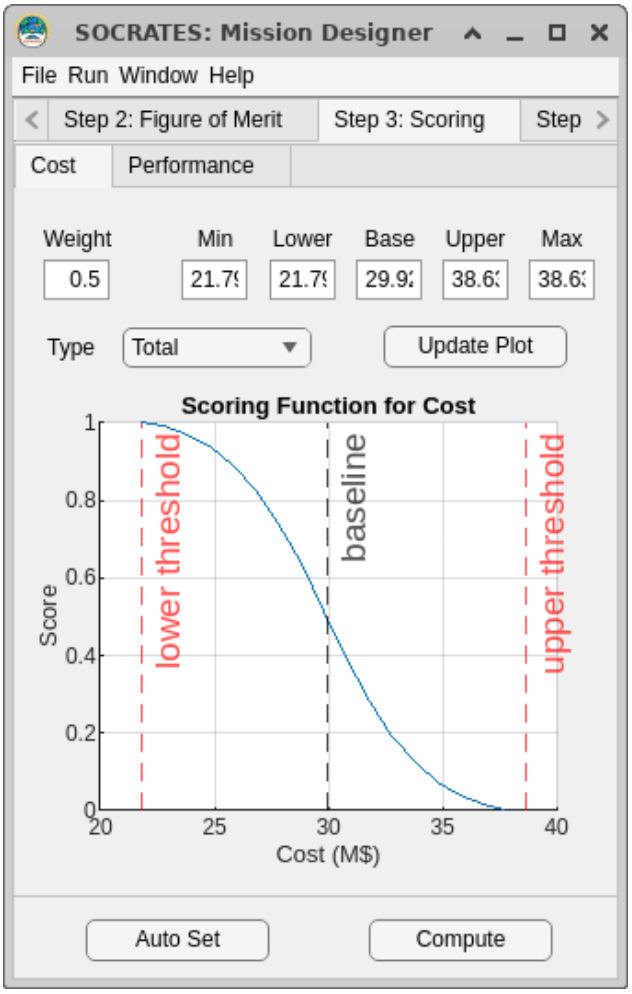



Figure 6.10: Scoring tab of the mission design tool.

Follow each of the steps below to set parameters for score mapping and compute the final scores for design alternatives.

1. **Configuring Weights**

   Set each of the weights of FoM metrics. The sum of weights for the same layer must be 1. The weight fields on the top-left corner of the Cost tab and the Performance tab set the weights for cost and performance in the top-level FoM structure. The weights for performance metrics can be set in each metric tab under the Performance tab. To set the weights for transmitter combinations and frequency combinations, select each node and click the "Edit" button to open the Scoring Function Editor shown in Figure 6.11. In the Scoring Function Editor, the user can set the weight for each transmitter/frequency combination.

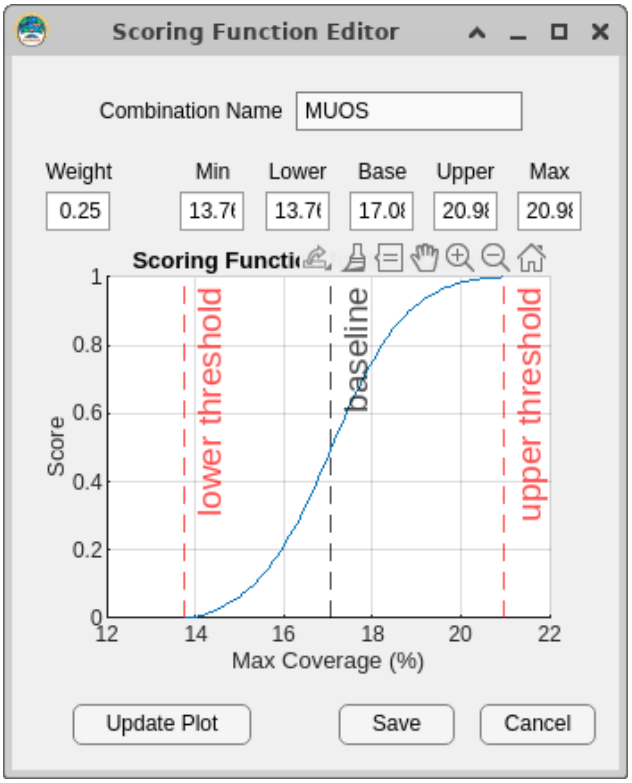



Figure 6.11: Scoring Function Editor.

2. **Configuring FoM Types**

   Select the type of FoM metrics used in the score calculation from the dropdown menu in each metric's tab. In the MSC tab, there are four types of metrics:

   - 1T visit: MSC based on the 1st access of the multi-frequency observation using the observation time window defined in the Figure of Merit tab. It is a 12-hour window in this demo project.
   - 1T revisit: MSC based on the 2nd access of the multi-frequency observation using the observation time window defined in the Figure of Merit tab. It is a 12-hour window in this demo project.
   - 2T visit: MSC based on the 1st access of the multi-frequency observation using an observation time window twice the length of the observation time window defined in the Figure of Merit tab. It is a 24-hour window in this demo project.
   - 2T revisit: MSC based on the 2nd access of the multi-frequency observation using an observation time window twice the length of the observation time window defined in the Figure of Merit tab. It is a 24-hour window in this demo project.

   In the MTC tab, there are two types of metrics:

   - 1T global: Globally averaged MTC based on the observation time window defined in the Figure of Merit tab. It is a 12-hour window in this demo project.
   - 2T global: Globally averaged MTC based on an observation time window twice the length of the observation time window defined in the Figure of Merit tab. It is a 24-hour window in this demo project.

3. **Configuring Scoring Functions**

   Set parameters for defining the shape of the scoring function. The parameters include:

   - Baseline: Baseline requirement for the FoM metric. This sets the midpoint (0.5).

- Lower threshold: This sets the lowest point (0) for SSF3 and the highest point (1) for SSF9.
- Upper threshold: This sets the highest point (1) for SSF3 and the lowest point (0) for SSF9.

Note that a penalty of -10 is applied if a FoM value violates the lower threshold of SSF3 or the upper threshold of SSF9. These parameters can be set in the Cost tab and the Scoring Function Editor. For a quick and easy setting, click the "Auto Set" button to automatically set up the parameters of scoring functions for all FoM metrics. This sets the baseline, lower threshold, and upper threshold of the scoring function with the mean, minimum, and maximum values of the FoM.

4. **Computing Scores**

   Click the "Compute" button in the Scoring tab. It calculates the score for all design alternatives.

### 6.5.1 Output

The scores are stored in a MAT file:

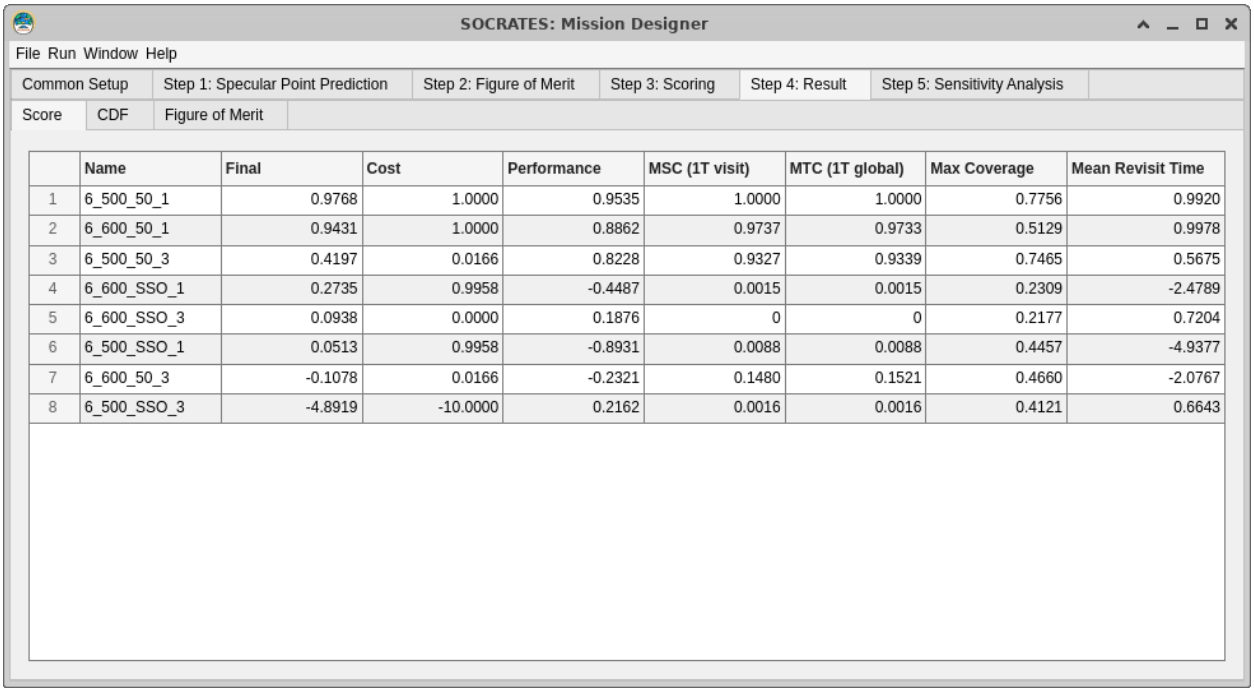
**./sim/Demo-design/out/score_Demo-design.mat**

This file will be used in the Result tab and the Sensitivity Analysis tab.

## 6.6 Result

This step loads the FoM MAT file from Section 6.4 and the score MAT file from Section 6.5 and displays the following:

- Design alternatives ranked by their final scores;
- Cumulative distribution function (CDF) of FoM;
- FoM values.




| | Name | Final | Cost | Performance | MSC (1T visit) | MTC (1T global) | Max Coverage | Mean Revisit Time |
|---|---|---|---|---|---|---|---|---|
| 1 | 6_500_50_1 | 0.9768 | 1.0000 | 0.9535 | 1.0000 | 1.0000 | 0.7756 | 0.9920 |
| 2 | 6_600_50_1 | 0.9431 | 1.0000 | 0.8862 | 0.9737 | 0.9733 | 0.5129 | 0.9978 |
| 3 | 6_500_50_3 | 0.4197 | 0.0166 | 0.8228 | 0.9327 | 0.9339 | 0.7465 | 0.5675 |
| 4 | 6_600_SSO_1 | 0.2735 | 0.9958 | -0.4487 | 0.0015 | 0.0015 | 0.2309 | -2.4789 |
| 5 | 6_600_SSO_3 | 0.0938 | 0.0000 | 0.1876 | 0 | 0 | 0.2177 | 0.7204 |
| 6 | 6_500_SSO_1 | 0.0513 | 0.9958 | -0.8931 | 0.0088 | 0.0088 | 0.4457 | -4.9377 |
| 7 | 6_600_50_3 | -0.1078 | 0.0166 | -0.2321 | 0.1480 | 0.1521 | 0.4660 | -2.0767 |
| 8 | 6_500_SSO_3 | -4.8919 | -10.0000 | 0.2162 | 0.0016 | 0.0016 | 0.4121 | 0.6643 |

Figure 6.12: Score tab in the Result tab of the mission design tool.

Figure 6.12 shows the Score tab listing the design alternatives ranked by the final scores. It also displays scores for cost, performance, MSC, MTC, maximum coverage, and revisit time. The user can resize the window of the mission design tool by dragging the edges of the window. Note that this demo project is intended to demonstrate the functionality of the mission design tool. For a design study or a tradespace analysis, the user should carefully configure the settings for simulation. For example, the simulation time should be long enough to represent the coverage of design alternatives. It is important to note that high scores mean that these are the preferred choices. Differentiation between design alternatives with close scores is ultimately up to the user. For this, the user can utilize the Sensitivity Analysis tab in the next step.

The CDF and the values of all FoMs can be displayed in the CDF tab and the Figure of Merit tab, respectively, as shown in Figure 6.13. For plotting the CDF, select the FoM of interest and the transmitter combination or the frequency combination from the dropdown menus and click the "Plot" button. The CDF is useful for determining the distribution of FoMs and adjusting the values of parameters for scoring functions if needed.

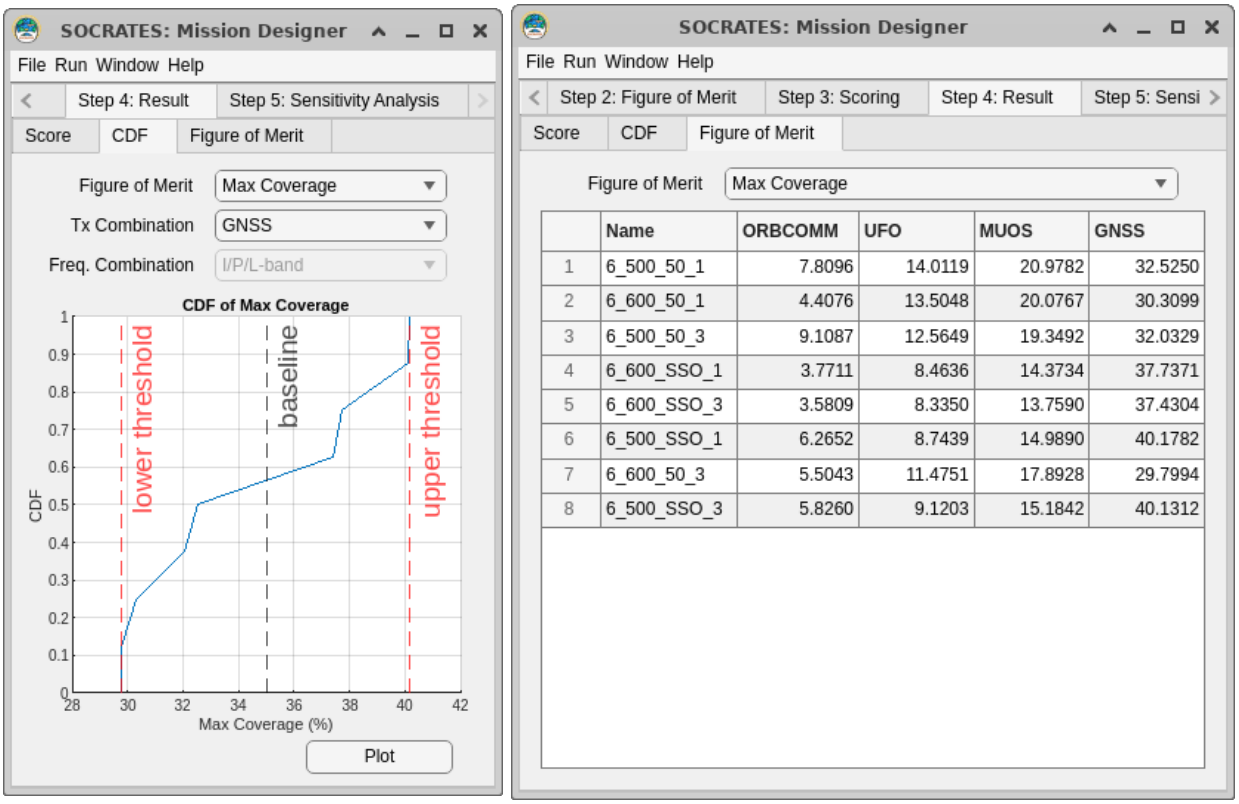



Figure 6.13: (Left) CDF tab and (right) Figure of Merit tab in the Result tab of the mission design tool.

## 6.7 Sensitivity Analysis

This step provides the user an environment to explore the tradespace of multi-SoOp constellations. Figure 6.14 shows the example of the scatter plot and the box plot. The user can resize the window of the mission design tool by dragging the edges of the window. For plotting the 2-D scatter plot, select X data and Y data from the dropdown menus, set Z data to "None", and click the "Plot" button. For plotting the 3-D scatter plot, select X, Y, and Z data from the dropdown menus and click the "Plot" button. Select the transmitter combination and/or the frequency combination accordingly. The color code provides additional information to the scatter plot. The user can also select specific values of design variables and filter out design alternatives that do not use those values. For this, use the dropdown menus in the last two rows.

For plotting the box plot, select X data (design variable) and Y data and click the "Plot" button. The user can also select specific values of design variables and filter out design alternatives that do not use those

values. For this, use the dropdown menus in the last two rows.

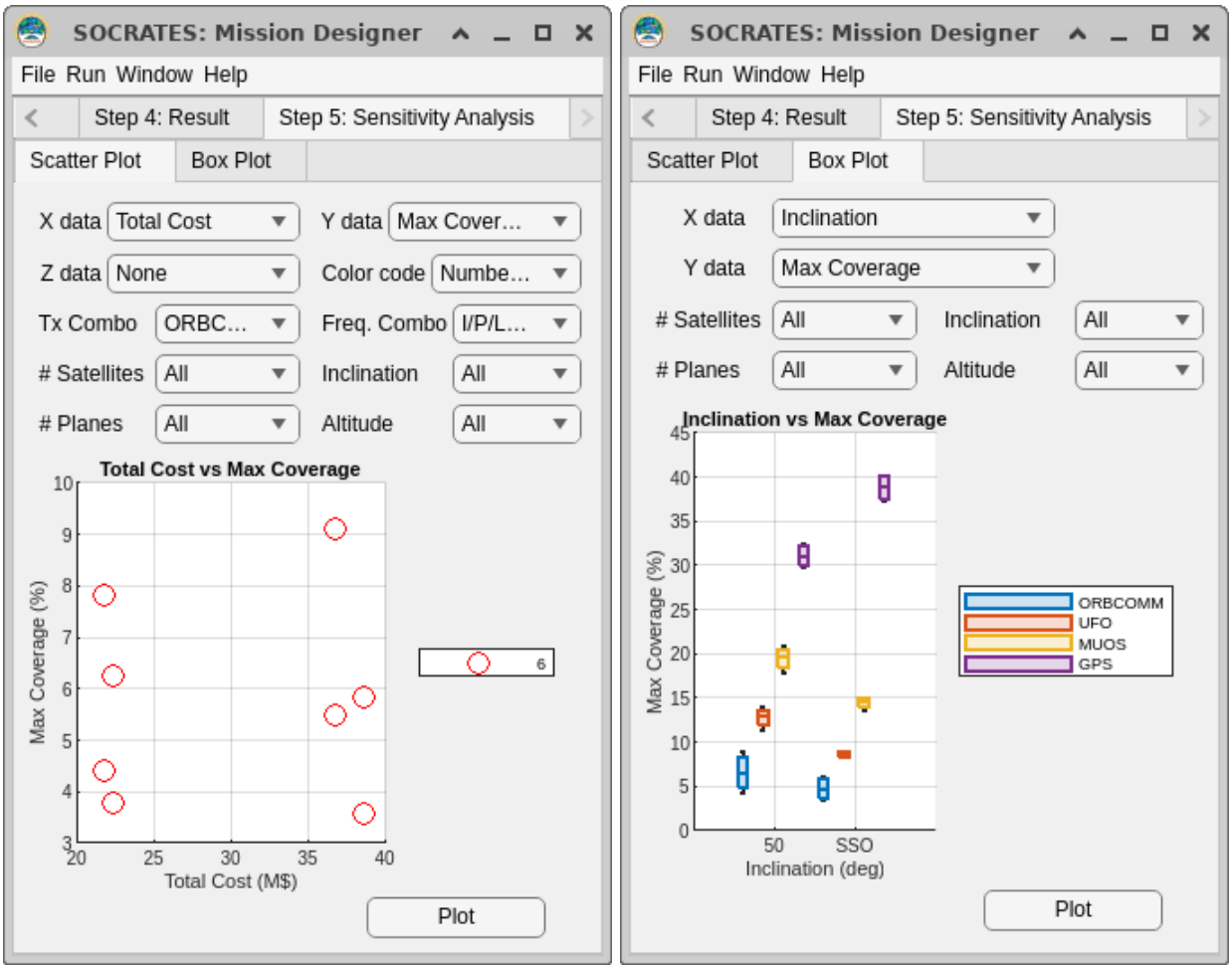



Figure 6.14: (Left) Scatter Plot tab and (right) Box Plot tab in the Sensitivity Analysis tab of the mission design tool.

# Chapter 7

# Input Files

## 7.1 Inp_Main.txt

This file is the top-level input file that describes common input parameters for all analysis types:

- `Simulation Mode`: `FIXED/INORBIT/DESIGN` selects the simulation project and corresponding input parameters. Input variables in the unselected mode are ignored.
- `Dynamic Data`: `TRUE/FALSE` turns on/off the corresponding dynamic data: soil moisture, soil temperature, freeze/thaw state, snow cover, and ground structure.
- `Single-Layered Soil`: Input parameters for single-layered ground structure. Skip if the multi-layered ground structure is selected.
  - `VSM File Name`: A file name for volumetric soil moisture values. If the fixed observation is selected, a text file should be provided. If satellite-based observation, a binary file converted from 1624×3856 32-bit floating-point data should be used. This file should be located in **./earthdata/dynamic** folder.
  - `Soil Temperature File Name`: A file name for soil temperature. This file should be located in **./earthdata/static** folder.
  - `RMSH Roughness of Soil Surface`: Root-Mean-Square-Height roughness of soil surface in centimeters.
- `Multi-Layered Soil`: Input parameters for multi-layered ground structure. Skip if the single-layered ground structure is selected.
  - `Number of Input Soil Layers`: Set the number of soil layers.
  - `Layer #`: This set of parameters must be repeated as many times as the number of input soil layers the user has set above.
    - `Depth of Soil Layer #`: Set the depth of corresponding soil layer in centimeters.
    - `VSM File Name Layer #`: A file name for volumetric soil moisture values. If the fixed observation is selected, a text file should be provided. If satellite-based observation, a binary file converted from 1624×3856 32-bit floating-point data. This file should be located in **./earthdata/dynamic** folder.
  - `Other Parameters`
    - `Air Layer Thickness`: Set air layer thickness in centimeters.
    - `Bottom-most Layer Thickness`: Set the bottom-most layer thickness in meters. This should generally be defined to be greater than or equal to the penetration depth of the SoOp transmitter's frequency and beneath the deepest point of interest within the soil moisture profile. The total layer depth will then become the largest value of `Depth of Soil Layer` plus this value.

    * `Layer Discretization`: Set layer discretization in meters. Only one value is enough to make discretization through all the ground layers.
    * `Fit Func for Diel. Profile`: `POLY2ND` / `POLY3RD` / `DISCRETE_SLAB` (not supported) / `LOGISTIC` sets a fitting function for a dielectric profile through the multiple ground layers.
    * `Soil Temperature File Name`: A file name for soil temperature. This file should be located in **./earthdata/dynamic** folder.

- `Snow Cover`: file names for percentage snow cover for 12 months. The MOD10CM products in 2019 are used as the default data. This file should be located in **./earthdata/dynamic** folder.

- `Static Data`: `TRUE`/`FALSE` turns on/off the corresponding static data.

- `Digital Elevation Model`

  – `DEM File Name`: A file name for digital elevation model. The default file is 1 arc-min Earth2014 model. This file should be located in **./earthdata/static** folder.

- `Land Type`

  – `Land Mask File Name`: A file name for land mask data. The default file is 34,704×14,616 global land mask on 1 km resolution EASE-Grid 2.0. This file should be located in **./earthdata/static** folder.

  – `Land Cover File Name`: A file name for land cover data. The default file is MODIS Land Cover MCD12C1 IGBP. This file should be located in **./earthdata/static** folder.

- `Soil Texture`

  – `(FIXED) File Name`: A file name that contains a set of soil texture data: sand fraction, clay fraction, and bulk density. It is only used in fixed observation. This file should be located in **./earthdata/static** folder.

  – `(INORBIT) Sand Ratio File Name`: A file name for mass fractions of sand. The default file is SMAP ancillary static data with a 1 km resolution. It is only used in satellite-based observation. This file should be located in **./earthdata/static** folder.

  – `(INORBIT) Clay Ratio File Name`: A file name for mass fractions of clay. The default file is SMAP ancillary static data with a 1 km resolution. It is only used in satellite-based observation. This file should be located in **./earthdata/static** folder.

  – `(INORBIT) Bulk Density File Name`: A file name for bulk density. The default file is SMAP ancillary static data with a 1 km resolution. It is only used in satellite-based observation. This file should be located in **./earthdata/static** folder.

- `In-situ Soil Moisture Network`: Optional input parameters to simulate the coverage of SNOOPI on cal/val sites.

- `MUOS Spotbeam`: Not used.

## 7.2 Inp_Inorbit.txt

This file describes input parameters for satellite-based and tower-based observation:

- `Number of Receivers`: The total number of receivers in a constellation (NRx).

- `Number of Transmitter Systems`: The total number of transmitter constellations (NSoOp). This number decides the number of a set of `SoOp Configuration #` and `42 Orbits` that follow. Note that this is not the number of satellites in a constellation.

- `Orbit Propagation Model`: `IMPORT`/`42` selects whether the simulator imports satellite states (position and velocity) from a user-specified text file or propagates using "42" simulator.

- `Import Orbits`: Parameters required if the orbit propagation model is chosen to be `IMPORT`. See Section 2.3.2.
  - `Number of Spacecrafts`: The total number of spacecraft, which is equal to the sum of the number of receivers and transmitters in all constellations.
  - `Number of Lines in Files`: The total number of sample states of satellites. All files must have the same length of data.
  - `SC # File Name`: A file name for satellite states: modified Julian day, position (m), and velocity (m/s) in Earth-centered inertia (ECI) coordinate system. The number of files must be the same as the number of spacecraft given above. Note that the satellite states files have headers in the first line. This file should be located in **./orbit/import** folder.
- `42 Orbits`: Parameters required if the orbit propagation model is chosen to be `42`. See Section 2.3.1.
  - `Rx Name, First RefOrb, Last RefOrb`: The receiver name is used to label the output file. The first and last reference orbit number specified in **Inp_Sim.txt** should be given. Receivers must be placed from orbit number 0 to orbit number (NRx-1).
  - `Tx # Name, First RefOrb, Last RefOrb`: The transmitter constellation name is used to label the output file. The first and last reference orbit number specified in **Inp_Sim.txt** should be given. Transmitters must be placed after all the receivers assigned to the orbits. This line must be repeated as many times as the number of transmitter systems (NSoOp) the user has set.
- `SoOp Configuration #`: Parameters required for a receiver to generate SoOp measurements with a corresponding transmitter system. This set of parameters must be repeated as many times as the number of transmitter systems (NSoOp) the user has set.
  - `Max Number of Specular Points`: The maximum number of specular points that will be predicted simultaneously. The specular points closest to the zenith of a receiver are selected first.
  - `Receive Antenna`
    - `Gain, Polarization`: The antenna gain of the receiver should be given in decibels. The polarization of the receiver can be chosen to be: `R`: RHCP, `L`: LHCP, `X`: linear vertical polarization, `Y`: linear horizontal polarization.
    - `Antenna Pattern Input`: `IDEAL`/`USER_DEFINED` selects the receiver antenna pattern.
    - `(USER_DEFINED) Antenna Pattern File Name`: If the receiver antenna pattern is selected to be `USER_DEFINED`, four Excel spreadsheets are required, which holds the normalized voltage values for co- and cross-polarizations for X and Y ports. Antenna patterns should be provided such that in each sheet of the Excel file, columns represent the theta-angle look-up and rows represent the phi-angle look-up. Theta ($\theta$) angles have a 180°scan, while phi ($\phi$) angles have a 360°scan. Note that the antenna pattern files do not have headers within the Excel spreadsheet. Each sheet consists of numbers only. This file should be located in **./ant_pat** folder.
  - `Transmitter`
    - `Frequency`: The transmitter antenna's operating frequency in MHz units.
    - `EIRP, Polarization`: EIRP of the transmitter should be given in decibels. The polarization of the transmitter can be chosen to be: `R`: RHCP, `L`: LHCP, `X`: linear vertical polarization, `Y`: linear horizontal polarization.
    - `Bandwidth of channel`: The bandwidth of transmitter's channel in kHz units. (Not used)
  - `Instrument`
    - `Coherent Integration Time`: Coherent integration time in seconds. It is used to calculate the signal-to-noise ratio.
    - `Noise Temperature Sky-view, Earth-view`: Noise temperature of the receiver in Kelvin. Sky-view noise temperature is used to calculate the SNR of the direct signal. Earth-view noise temperature is used to calculate the SNR of the reflected signal.

- Other Parameters
  - Minimum Local Surface Resolution: Once an initial specular point is determined, a local grid of points is constructed to account for the local elevation. This parameter defines the angular increments in theta direction of the traditional spherical coordinates defined at a local surface frame.
  - Number of Grid Points in theta, phi direction: The number of grid points in theta defines the number of angular increments in theta direction. The number of grid points in the phi direction defines the angular increments in the phi direction.
  - Gain of Gradient Function: Gain for the specular point determination algorithm. Adjust this value only if the user receives an error message "Exceed max iteration ...". Rx and Gs(Not used) are the gain for space-based and ground-based observation, respectively.

## 7.3 Inp_Fixed.txt

This file describes input parameters for fixed observation:

- Receiver Altitude: The altitude of a receiver in kilometers. The altitude makes only the difference on the received power calculations.
- Receive Antenna Gain, Polarization: The antenna gain of the receiver should be given in decibels. The polarization of the receiver can be chosen to be: R: RHCP, L: LHCP, X: linear vertical polarization, Y: linear horizontal polarization.
- Antenna Pattern Input: IDEAL/USER_DEFINED selects the receiver antenna pattern.
- (USER_DEFINED) Antenna Pattern File Name: If the receiver antenna pattern is selected to be USER_DEFINED, four Excel spreadsheets are required, which holds the normalized voltage values for co- and cross-polarizations for X and Y ports. Antenna patterns should be provided such that in each sheet of the Excel file, columns represent the theta-angle look-up and rows represent the phi-angle look-up. Theta ($\theta$) angles have a 180°scan, while phi ($\phi$) angles have a 360°scan. Note that the antenna pattern files do not have headers within the Excel spreadsheet. Each sheet consists of numbers only. This file should be located in **./ant_pat** folder.
- Receiver Elevation, Azimuth: The elevation ($\theta$) and azimuth ($\phi$) of the receiver orientation in degrees. The elevation complements the incidence angle to 90 degrees, i.e. 90-$\theta$. This is the angle formed between the ground surface and the vector located between the specular point along the ground surface and the receiver. The azimuth is the heading or compass direction described clockwise from magnetic North.
- Transmitter Frequency: The transmitter antenna's operating frequency in MHz units.
- Transmitter Range to Earth Center: The transmitter's range to Earth's center should be given in kilometers.
- Transmitter EIRP, Polarization: EIRP of the transmitter should be given in decibels. The polarization of the transmitter can be chosen to be: R: RHCP, L: LHCP, X: linear vertical polarization, Y: linear horizontal polarization.
- Transmitter Elevation, Azimuth: The elevation ($\theta$) and azimuth ($\phi$) of the transmitter orientation in degrees. The elevation complements the incidence angle to 90 degrees, i.e. 90-$\theta$. This is the angle formed between the ground surface and the vector located between the specular point along the ground surface and the transmitter. The azimuth is the heading or compass direction described clockwise from magnetic North.

## 7.4 Inp_Report.txt

This file lists available outputs from satellite-based observation, tower-based observation, and mission design. The user can turn on and off each output parameter with `TRUE` and `FALSE`. This file can be updated by using the GUI. See Chapter 8 for more information.

## 7.5 (42) Inp_Sim.txt

- `Simulation Control`
  - `Time Mode`: The default is `FAST`.
  - `Sim Duration, Step Size`: Simulation duration will determine the simulated mission lifetime. Step size is a discrete-time step in numerical propagation.
  - `File Output Interval`: It will determine the time interval of output generation specified in **Inp_Report.txt**.
  - `Graphics Front End?`: The graphic user interface is not supported.
  - `Command Script File Name`: The attitude command can be generated through this script. **Inp_Cmd.txt** provides examples of command formats.
- `Reference Orbits`
  - `Number of Reference Orbits`: The number of orbits that will be used in the simulation. A constellation may have multiple orbits.
  - `Input file name for Orb #`: `TRUE`/`FALSE` turns on/off the corresponding orbit #. The orbit input files are located in **./orbit/42input/Orb**. This line must be repeated as many times as the number of reference orbits the user has set. Place the orbits of receivers from the top followed by transmitters.
- `Spacecraft`
  - `Number of Spacecraft`: The number of spacecraft that will be used in the simulation.
  - `Existence, RefOrb, Input file for SC #`: `TRUE`/`FALSE` turns on/off the corresponding spacecraft #. The spacecraft input files are located in **./orbit/42input/SC**. This line must be repeated as many times as the number of spacecraft the user has set. Place the spacecraft of receivers from the top followed by transmitters.
- `Environment`: `Date` and `Time` determine the starting epoch of the simulation.
- `Ground Stations`: Not used.

# Chapter 8

# Output Files

## 8.1 SoOp-related Output

The output of the SoOp observation for each receiver is written in [**Receiver constellation name**]#_[**Transmitter constellation name**]**.out** in the **./sim/[project name]/out** folder, in which # is the receiver satellite number. The [**Receiver constellation name**] and [**Transmitter constellation name**] are the string inputs from **Inp_Inorbit.txt**. In GUI, they are the name of the constellation node. The outputs are generated in every file output interval specified in **Inp_Sim.txt**. Note that these outputs are only available when a specular point between a receiver and a transmitter does exist at a time instant. The outputs are separated by a space in a single line and are printed in the order in **Inp_Report.txt**. Table 8.1 summarizes the output parameters for SoOp observation.

Table 8.1: Summary of SoOp-related output parameters

| Parameter | Description | Unit | Column Size | Available from |
|---|---|---|---|---|
| Julian Day | Julian date for the specular point. | UTC | 1 | Space&Tower |
| Date | Year, month, day, hour, minute, and second for the specular point. | UTC | 6 | Space&Tower |
| Rx LLA | Latitude, longitude, and altitude above the WGS84 ellipsoid of the receiver satetlite. | deg, deg, m | 3 | Space&Tower |
| Rx Pos ECEF | Cartesian position of the receiver in the Earth-centered, Earth-fixed coordinate system. | meters | 3 | Space&Tower |
| Rx Vel ECEF | Cartesian velocity of the receiver in the Earth-centered, Earth-fixed coordinate system. | m/s | 3 | Space&Tower |
| Tx ID | A number identifying which transmitter the specular point originates from. The ID corresponds to the order of the transmitter satellites in **Inp_Sim.txt**. | numeric | 1 | Space&Tower |
| Tx LLA | Latitude, longitude, and altitude above the WGS84 ellipsoid of the transmitter satellite. | deg, deg, meters | 3 | Space&Tower |
| Tx Pos ECEF | Cartesian position of the transmitter in the Earth-centered, Earth-fixed coordinate system. | meters | 3 | Space&Tower |

| | | | | |
|---|---|---|---|---|
| Tx Vel ECEF | Cartesian velocity of the transmitter in the Earth-centered, Earth-fixed coordinate system. | m/s | 3 | Space&Tower |
| Sp LLA | Latitude, longitude, and altitude above the WGS84 ellipsoid of the specular point. | deg, deg, m | 3 | Space&Tower |
| Sp Pos ECEF | Cartesian position of the specular point in the Earth-centered, Earth-fixed coordinate system. | meters | 3 | Space&Tower |
| LoS Direct | Azimuth and elevation angles of the incoming direct signal to the receiver satellite. | deg | 2 | Space&Tower |
| LoS Reflect | Azimuth and elevation angles of the incoming reflected signal to the receiver satellite. | deg | 2 | Space&Tower |
| Delay Doppler | Delay and Doppler shift of the direct signal, delay and Doppler shift of the reflected signal observed by the receiver. | sec, Hz, sec, Hz | 4 | Space&Tower |
| Ant Gain Direct | Antenna gain of the receiver for the direct signal. | dB | 1 | Space&Tower |
| Ant Gain Reflect | Antenna gain of the receiver for the reflected signal. | dB | 1 | Space&Tower |
| FFZ | Semi-major axis, semi-minor axis, and azimuth angle of the first Fresnel zone. | m, m, deg | 3 | Space&Tower |
| Incidence Angle | Incidence angle of the reflected signal. | deg | 1 | Space&Tower |
| Elevation | Elevation angle of the reflected signal. | deg | 1 | Space&Tower |
| RPY to Sp | Roll, pitch, and yaw in 1-2-3 rotations from local-nadir to the specular point. | deg | 3 | Space |
| Land mask | A indicator with a '1' where the specular point is over land and a '0' where it is over water. | numeric | 1 | Space |
| VSM | $N$-layered volumetric soil moisture at the specular point. | $m^3/m^3$ | $N$ | Space |
| Soil Temperature | Soil temperature at the specular point. | °C | 1 | Space |
| Freeze Thaw | A indicator with a '1' where the specular point is over frozen ground and a '0' where it is not. | numeric | 1 | Space |
| Soil Texture | Sand ratio, clay ratio, and soil bulk density at the specular point. | num, num, $g/cm^3$ | 3 | Space |
| Snow Cover | Fraction of snow coverage at the specular point. | % | 1 | Space |
| Refl Coeff Co | Real and imaginary parts of the reflection coefficient at co-polarization. | numeric | 2 | Space |
| Refl Coeff X | Real and imaginary parts of the reflection coefficient at cross-polarization. | numeric | 2 | Space |
| Refl Co | Reflectivity and reflectivity with error at co-polarization. | numeric | 2 | Space |

| | | | | |
|---|---|---|---|---|
| Refl X | Reflectivity and reflectivity with error at cross-polarization. | numeric | 2 | Space |
| SNR Direct | Co-polarized and cross-polarized signal-to-noise ratios of the direct signal. | dB | 2 | Space |
| SNR Reflect | Co-polarized and cross-polarized signal-to-noise ratios of the reflected signal. | dB | 2 | Space |
| MUOS Spotbeam | IDs of the MUOS spotbeam where the receiver and the specular point are located. | numeric | 2 | Space |

## 8.2 Subsystem Output

The subsystem outputs are generated in separate files from the SoOp-related output files. Each selected output is written in every file output interval specified in **Inp_Sim.txt**. Table 8.2 summarizes the parameters and their files for subsystem outputs.

Table 8.2: Summary of subsystem output parameters

| Subsystem | Parameter | | Description |
|---|---|---|---|
| Comm. | Access Time/Arc | | This creates two files: **AccessTime_Rx#.out** and **AccessArc_Rx#.out**. **AccessTime_Rx#.out** contains the ground station ID (specified in **Inp_Subsys.txt**), access count, start time (in seconds), and end time (in seconds). **AccessArc_Rx#.out** contains the ground station ID, access count, latitude (in deg), longitude (in deg), and altitude (in meters) of the receiver. |
| ADCS | Truth | Julian Day | Julian date for the output. Included in **TrueAttitude_Rx#.out**. |
| | | Pos ECI | Cartesian position of the receiver in the Earth-centered, Earth-fixed coordinate system. Included in **TrueAttitude_Rx#.out**. |
| | | RPY | Roll, pitch, yaw (in degrees) of the receiver satellite. Included in **TrueAttitude_Rx#.out**. |
| | | Ang Vel | Cartesian angular velocity (in degrees per second) of the receiver satellite. Included in **TrueAttitude_Rx#.out**. |
| | | Ang Mom | Cartesian angular momentum (in Netwon-meter-second) of the receiver satellite.Included in **TrueAttitude_Rx#.out**. |
| | | Mag Field | Cartesian magnetic field (in Tesla) at the location of the receiver satellite. Included in **TrueAttitude_Rx#.out**. |
| | | Eclipse Flag | A indicator with a '1' where the receiver satellite is in eclipse and a '0' where it is not. Included in **TrueAttitude_Rx#.out**. |

| | | |
|---|---|---|
| Sensor | Gyros | This creates a file named **Gyro_Rx#.out** containing data from $N$ gyro sensors. The first three columns contain the Cartesian angular velocity (in degrees per second) of the receiver satellite. The next $N$ columns contain the true rates (in deg/sec) of each of the $N$ gyro sensors. The next $N$ columns contain the measured rate (in deg/sec) of each sensor. The next $N$ columns contain the bias (in deg/sec) of each sensor. The next $N$ columns contain the angle measurement (in deg) for each sensor. |
| | Magnetometers | This creates a file named **Magnetometer_Rx#.out** containing data from $N$ magnetometers. The first three columns contain the Cartesian magnetic field (in Tesla) at the location of the receiver satellite. The next $N$ columns contain the measurement of each of the $N$ magnetometers. |
| | Accelerometers | This creates a file named **Accel_Rx#.out** containing data from $N$ accelerometers. The first $N$ columns contain the true acceleration (in $m/s^2$) of each of the $N$ accelerometers. The next $N$ columns contain the measurement of each of the $N$ accelerometers. |
| | Reaction Wheels | This creates a file named **Wheel_Rx#.out** containing data from $N$ reaction wheels. The first three columns contain the Cartesian torque command to the receiver satellite. The next $N$ columns contain the exerted torque on each of the $N$ reaction wheels. The next $N$ columns contain the angular momentum (in Newton-meter-second) of each reaction wheel. |
| | Magnetorquers | This creates a file named **Magnetorquer_Rx#.out** containing data from $N$ magnetorquers. The first three columns contain the Cartesian magnetic dipole command to the receiver satellite. The next $N$ columns contain the exerted torque by each of the $N$ magnetorquers. The next $N$ columns contain the magnetic dipole of each magnetorquer. |

# References


[1] Eric Stoneking. 42: A General-Purpose Spacecraft Simulation. NASA Software Designation GSC-16720-1, 2010–2019.

[2] Mary J Brodzik, Brendan Billingsley, Terry Haran, Bruce Raup, and Matthew H Savoie. Ease-grid 2.0: Incremental but significant improvements for earth-gridded data sets. *ISPRS International Journal of Geo-Information*, 1(1):32–45, 2012.

[3] Mehmet Kurum, Manohar Deshpande, Alicia T Joseph, Peggy E O'Neill, Roger H Lang, and Orhan Eroglu. Scobi-veg: A generalized bistatic scattering model of reflectometry from vegetation for signals of opportunity applications. *IEEE Transactions on Geoscience and Remote Sensing*, 57(2):1049–1068, 2018.